

Advanced surface wave methods

The most common surface wave testing on land consists of the acquisition of vertical component Rayleigh wave data and the inversion of the dispersion to obtain a shear wave velocity model. The extension to the estimation of dissipative properties has been discussed in previous chapters.

In this chapter, we present some alternative approaches and applications. The objective is not to provide an exhaustive overview of the recent surface wave research but rather to provide an insight into advanced topics and further developments. Analogies and differences with respect to the standard surface wave testing are also discussed.

The following topics will be presented in particular: the use of Love waves on land and Scholte waves in marine applications; joint inversion of surface wave data with other geophysical measurements, with examples on the joint inversion of critically refracted body waves and surface waves; passive seismic interferometry; surface wave polarization; and passive techniques based on the analysis of the horizontal-to-vertical spectral ratio.

8.1 LOVE WAVES

The existence and properties of surface waves characterized by pure horizontal motion were first discussed by Love (1911). Therefore, these surface waves are called Love waves. They propagate as a pure shear disturbance; the particle displacement is horizontal, parallel to the free surface, and perpendicular to the direction of propagation, as shown in Figure 8.1. The Love wave propagation velocity and attenuation depend on the shear properties of the subsurface. Love waves can be acquired, processed, and inverted with methods that are similar to those used for Rayleigh waves.

In the next section, the basic properties of the Love waves are summarized, the experimental configuration to be used for the acquisition is discussed, and an example of joint Love and Rayleigh wave testing is presented.

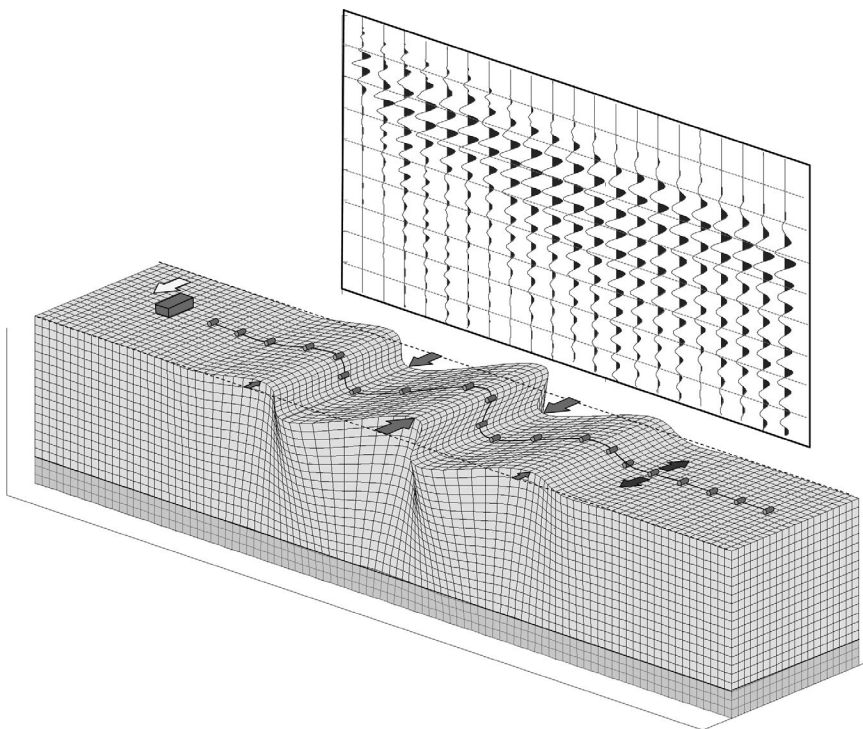


Figure 8.1 Schematic representation of Love wave propagation and their acquisition with horizontal sources and receivers on the ground surface.

8.1.1 The nature of Love waves

Love waves are surface waves containing only Shear, Horizontally polarised (SH) motion, while Rayleigh waves have coupled P and SV potentials (see Chapter 2). Although Rayleigh waves exist in a homogeneous half-space because of the interference between evanescent P -waves and phase shifted SV -waves, the Love waves do not exist in a homogeneous half-space.

The simplest model in which Love waves can exist is a low-velocity layer over a stiff half-space. The supercritical energy, impinging on the half-space with an angle larger than the critical angle, is totally reflected up and reaches the free surface where it is totally reflected down. Energy is then trapped in the low-velocity layer. In this case, the Love wave kinematics can be easily explained in terms of total internal reflections in the waveguide, interfering constructively and destructively as a function of the wavelength (Ewing et al. 1957).

Like Rayleigh waves, Love waves are modal and, in simple cases, they have simple modal shapes. In Figure 8.2, the displacement eigenfunctions

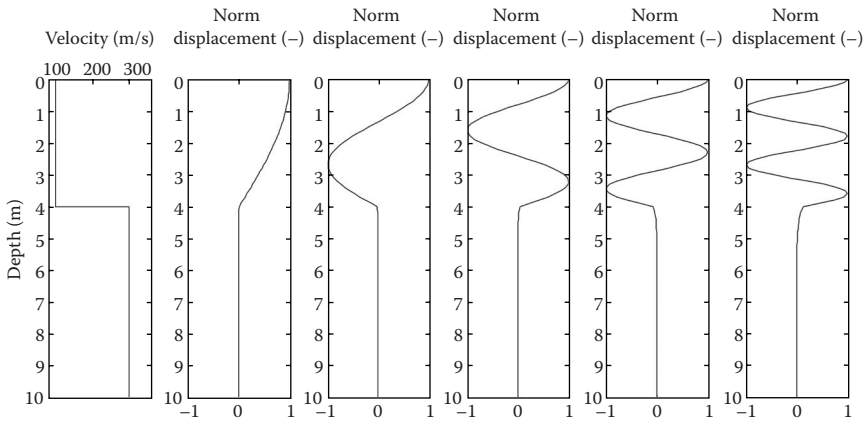


Figure 8.2 Displacement eigenfunctions of Love waves for the first five modes at 80 Hz for the shear wave velocity model reported in the panel on the left.

are plotted as a function of depth for the first five modes at the frequency of 80 Hz for a simple model of a single-layer waveguide (4 m with a shear wave velocity of 100 m/s) over a half-space (with a shear wave velocity of 300 m/s). The modal curves are plotted in Figure 8.3 and compared to the Rayleigh wave modal curves. Due to the high velocity contrast, the Rayleigh modes (Figure 8.3b) are more complex and show an osculation point between the fundamental mode and the first higher mode, indicated in the dashed box (Strobbia 2005).

The constructive interference of the SH-waves, within a limited range of incidence angles, creates resonance in the sedimentary layers. Usually, the SH resonances correspond to points of maximum curvature in the Love dispersion curves. The quarter wavelength rule often gives an acceptable estimate of the resonance frequency (i.e., the resonance frequency can be estimated as a quarter of the shear wave velocity of the top layer divided by its thickness).

This simple single-layer waveguide, however, is not the only case allowing the existence of Love waves. They exist in more complex models (e.g., in multilayer waveguides and with smoother velocity variations). In principle, Love waves also exist in the presence of a surface velocity inversion, with a low-velocity layer sandwiched between two stiffer layers. In practice, the generation and observation of Love waves can be difficult at sites with velocity inversions. The solution of the Love problem can be computed with similar approaches to the one used for Rayleigh waves (see Chapter 2).

The polarization of the shear component of the Rayleigh and of the Love waves is different—vertical for the Rayleigh wave, horizontal for the Love wave. In cases of material anisotropy, the Love wave and the Rayleigh wave are associated with the SH and the SV velocities, respectively.

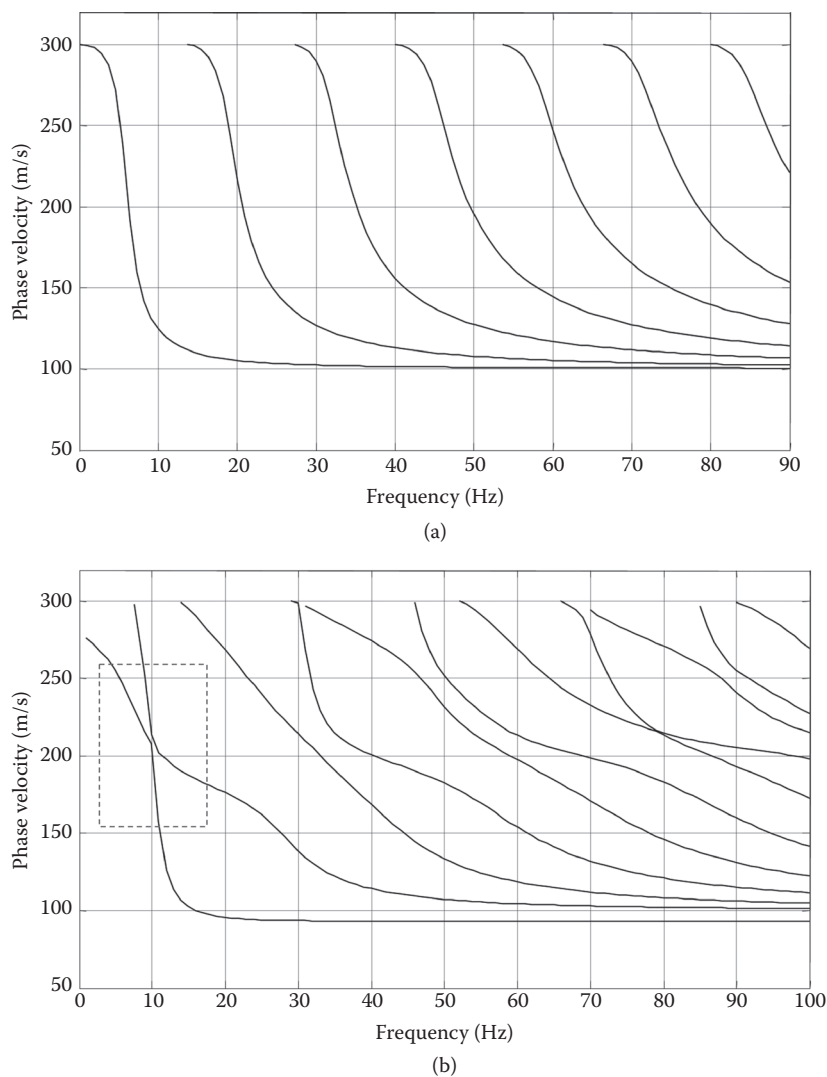


Figure 8.3 Modal curves for (a) Love and (b) Rayleigh modes for the simple model of Figure 8.2.

8.1.2 Experimental configurations

The particle displacement of the Rayleigh wave induced by a vertical force at the surface has a large vertical component at the free surface. Therefore, the acquisition of Rayleigh waves is straightforward with common engineering seismic equipment and with vertical source and receivers. On the contrary,

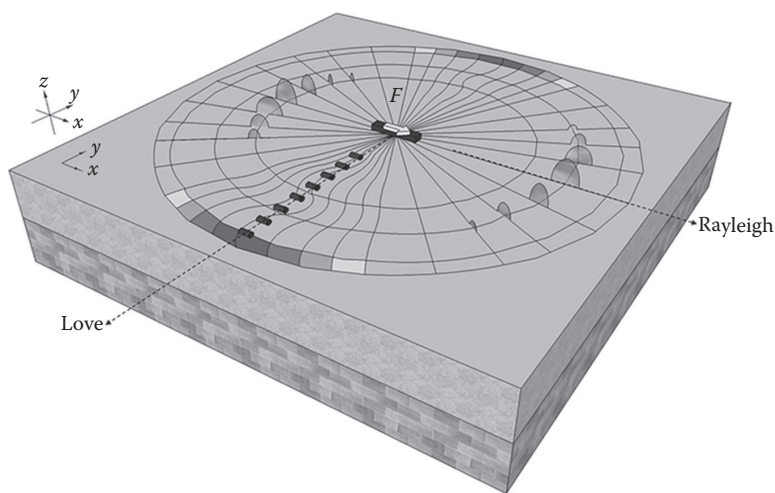
the Love wave displacement is horizontal and parallel to the direction of propagation; moreover, the largest Love wave radiation is from a horizontal source, in the direction perpendicular to the source force. A horizontal force is needed to generate Love waves. The wave propagation mainly consists of horizontal crossline particle motion. The horizontal force also generates Rayleigh waves, even if mainly in the direction of application of the force.

Considering a horizontal force in the x direction at the origin of the reference system xyz , the Love wave has a maximum displacement in the x direction, u_x , along the y -axis, where the x component of the horizontal motion of a Rayleigh wave is minimal. Figure 8.4 shows the radiation of a source at the surface, with the maximum Love wave in the perpendicular direction. The wiggle graph (Figure 8.4b) represents a snapshot of the horizontal displacement u_x ; along the x -axis, the contribution comes from the Rayleigh wave; along the y -axis, it comes from the Love wave.

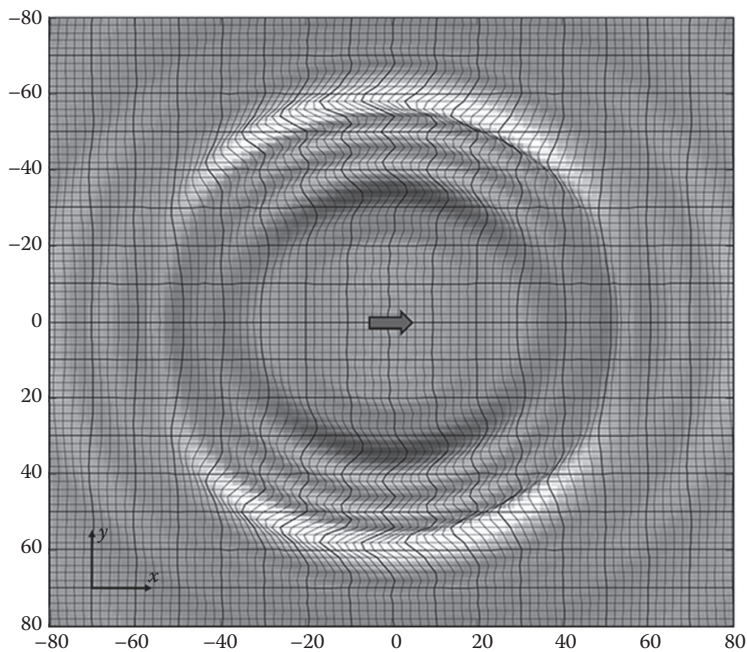
The general principles for the acquisition of Love waves are similar to those for Rayleigh waves, as described in Chapter 3. The design of the measurement of the wave field requires the same considerations about the effects of coherent and incoherent noise, about the consequences of the sampling, and about the field equipment and procedures. However, to excite Love waves, a horizontal source has to be used and horizontal receivers are required to detect the induced particle motion.

Even a perfectly horizontal source induces a pressure component, which will generate multiple events such as refractions and reflections, affecting the records even in the direction of maximum shear radiation. The shear component, SH body waves and Love waves, will be superimposed onto these other events, and it can be difficult to separate them. This is well known in the acquisition of SH body waves, for reflection and refraction surveying (Deidda and Balia 2001), for which specific acquisition procedures have been developed. For example, reversing the polarity of a horizontal source reverses the radiated shear polarity but not that of the pressure component; the subtraction of the corresponding horizontal component of the particle motion, therefore, subtracts the pressure component and sums the shear component. An efficient alternative to this field procedure is the use of specially coupled receivers that perform the synchronous subtraction electrically (Sambuelli et al. 2001), increasing the accuracy of the procedure and the operational efficiency.

As mentioned before, Love waves can have marked resonances in very shallow layers, producing narrow band long traces. The resonance frequency is a function of the layer properties and can be used to estimate the characteristics of the waveguide. Data affected by such resonances have a sharply peaked amplitude spectrum. However, despite the apparently narrowband spectral content, usually the experimental data can be processed to extract a dispersion curve over a wide frequency range. Figure 8.5 shows an experimental shot gather with strong Love waves, acquired with a horizontal source. Traces show long-resonating signals with



(a)



(b)

Figure 8.4 Radiation from a horizontal source (the arrows) with generation of Love waves and Rayleigh waves. (a) 3D view and (b) horizontal particle displacement in the x direction due to both Love and Rayleigh waves.

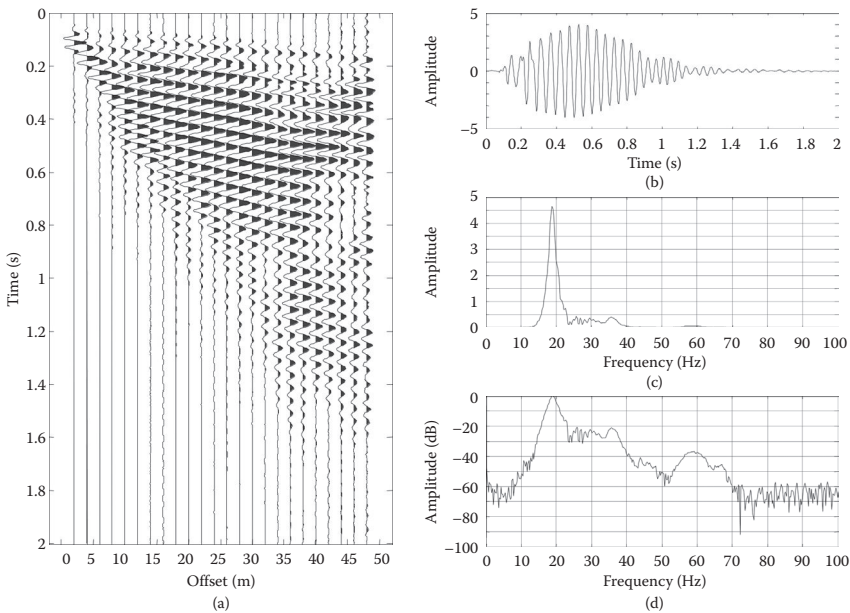


Figure 8.5 Example of Love wave experimental data: (a) shot gather; (b) example of a single trace (receiver offset equal to 30 m); (c) amplitude Fourier spectrum of the same trace in natural scale; (d) amplitude Fourier spectrum of the same trace in log scale.

a narrow spectrum at around 20 Hz. The traces at 30 m and its spectrum are plotted in Figures 8.5b and c. The representation of the spectrum in dB shows that at that offset the usable band is quite wide.

The data of Figure 8.5 are very dispersive as shown by the widening of the wave train, which reaches almost 2 s at 50 m offset. The experimental dispersion curve may be extracted with any of the methods reported in Chapter 4, and it is reported in Figure 8.6. The phase velocity varies from 50 to 500 m/s. The spectral peaks visible in Figure 8.5 (at about 19 Hz) correspond to the resonance of an extremely shallow layer, with a thickness of around 1 m and with a very large velocity contrast. The other peaks in the spectrum correspond to the resonance of the higher modes.

Rayleigh and Love waves' dispersion curves may be used for simultaneous inversion aimed at estimating the shear wave velocity profile under the assumption of isotropic behavior of the material.

Love waves can be acquired simultaneously to SH reflection and refraction data. The synergies between the techniques are similar to those between P-refraction and Rayleigh wave inversion, which will be discussed

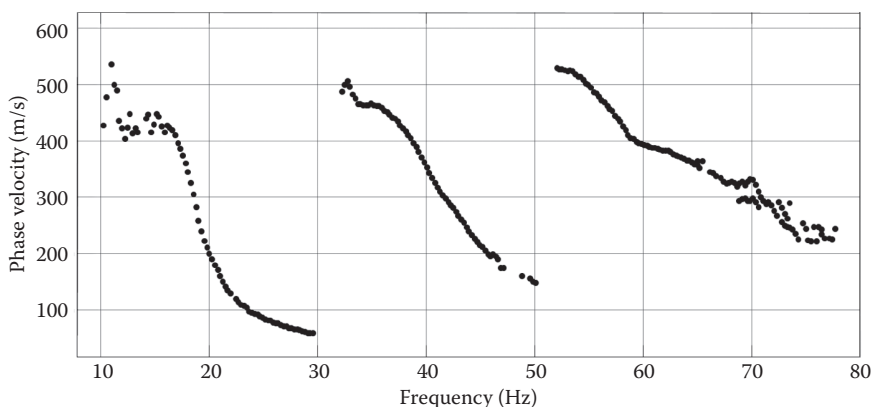


Figure 8.6 Experimental dispersion curves for Love waves extracted from the gather of Figure 8.5.

in Section 8.3. Body wave techniques may provide warnings about lateral variations and localize sharp boundaries, while surface waves overcome some limitations of the refraction (for instance, the hidden layer problem).

8.1.3 Real data example

A case history of acquisition of Love and Rayleigh waves is reported in the following. In particular, the availability of Love wave measurements allows some ambiguities in the Rayleigh wave dispersion extraction to be resolved.

The test site is located on Terceira Island in the Azores archipelago, in Sao Sebastiao, a village that suffered strong damage by an earthquake in 1980 due to local seismic site amplification. The village is located in a volcanic caldera, with basaltic bedrock and some lateral pyroclastic cones. As part of the site characterization, a geophysical campaign has been performed, acquiring surface waves (Rayleigh and Love) as well as refraction and reflection data (for P- and SH-waves) (Lopes et al. 2013).

Surface waves have been acquired with a multichannel seismograph with 24 active channels. For the Love wave acquisition, a wooden sleeper struck laterally using a sledgehammer has been used as source and horizontal swyphone geophones (Sambuelli et al. 2001) as receivers. Two persons were standing on the sleeper to increase the shear friction force on the ground. For the Rayleigh wave acquisition, a sledgehammer on steel plate as a seismic source has been used along with vertical 4.5 Hz geophones. The seismograms with the corresponding f - k spectra and dispersion curves are reported in Figures 8.7 and 8.8 for Rayleigh and Love waves, respectively.

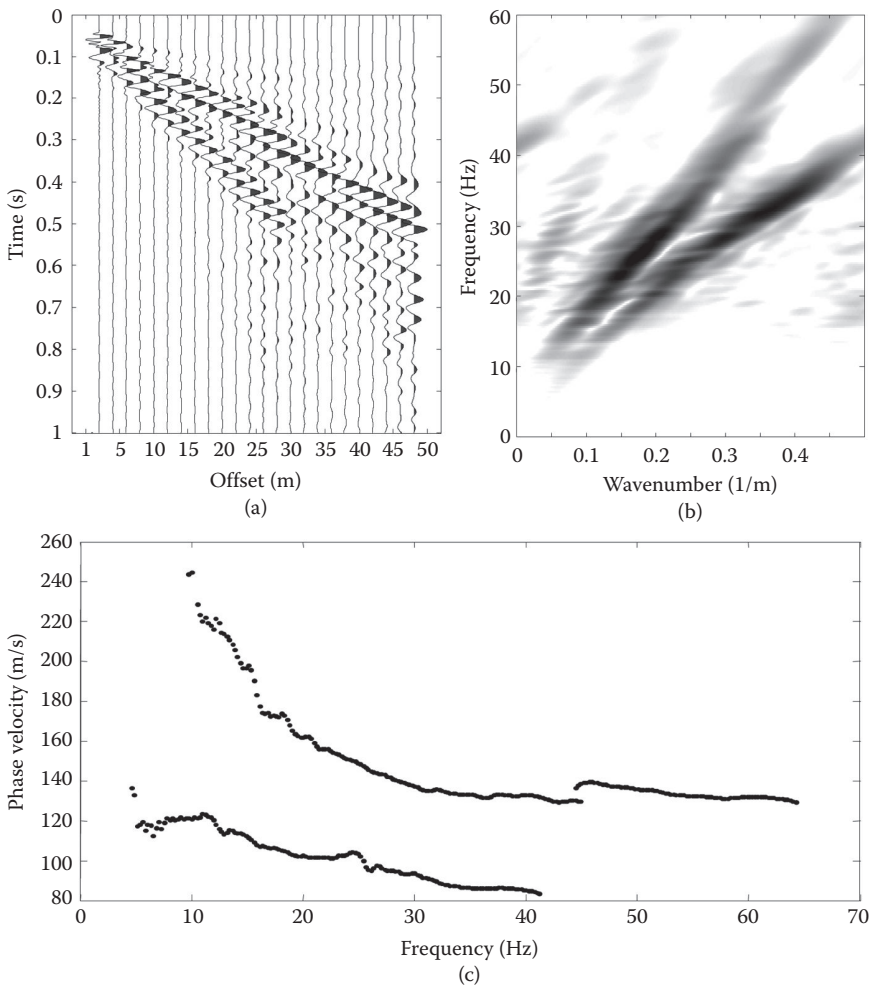


Figure 8.7 Rayleigh wave data: (a) seismogram; (b) frequency–wavenumber spectrum; (c) experimental curves.

The first Rayleigh mode suffers a strong attenuation and loses more than 1.3 dB/m (Figure 8.7). The higher mode, which is clearly visible in the t – x and f – k domains, is less attenuated because of the higher wavenumber. The Love wave gather shows a dominant fundamental mode with low velocity (Figure 8.8). The phase velocity of the Love wave first mode is better defined than the corresponding Rayleigh mode and confirms the velocity of the latter, resolving possible doubts about the Rayleigh modes.

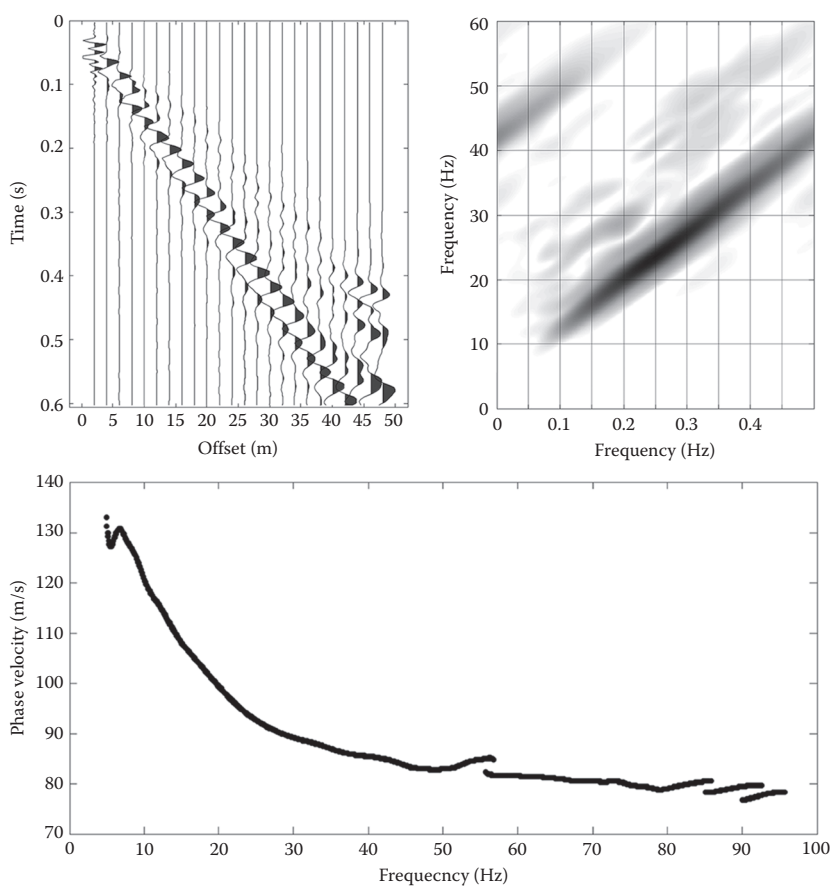


Figure 8.8 Love wave seismogram and dispersion.

8.2 OFFSHORE AND NEARSHORE SURFACE WAVE TESTING

The geophysical characterization of shallow soils and rocks under water is of great interest for a variety of applications, such as the design of port structures, offshore platforms, artificial islands, and sea bottom pipelines, and for environmental studies. The seismic parameters (velocity and attenuation) are needed for dynamic problems, for example, for the seismic site response assessment. They can also be useful in both offshore and nearshore applications for foundation design, slope stability evaluation, liquefaction potential assessment, and erosion potential studies. Moreover, the variations of properties in the shallow subseafloor are important in deep exploration

applications to compensate for the near-surface distortions (for example, with static corrections). The geophysical underwater applications are often called *marine* even if they are applied to lakes, rivers, and water supply reservoirs.

Specific high-frequency acoustic reflection systems and techniques are used to image, with a high resolution, the morphology of the sediments below the water bottom. For instance, sub-bottom profilers use sources of high-frequency pulses that are propagated through the water layer and are reflected by the seafloor and by the sub-bottom layers. Frequency-modulated pulses are used to optimize the trade-off between penetration and resolution. The result is a detailed geometrical characterization.

On the contrary, the geophysical in situ measurement of the mechanical properties is not straightforward, with the conventional shallow seismic techniques that are effectively applied onshore.

The P-wave velocity of soft, saturated sediments is dominated by the pore water compressibility. The variations of P-wave velocity are low and difficult to detect with conventional refraction or reflection methods. The conventional shallow P-wave refraction can be used to detect major boundaries, such as the interface between a stiff bedrock and the sediments above. Yet, it offers limited potential to discriminate among loose sediments. Shear wave measurements play a major role in the characterization of saturated sediments. Indeed, the shear wave velocity is controlled by the sediment solid skeleton and has a stronger correlation with geotechnical parameters. Furthermore, the shear wave velocity can be much lower than the compression velocity in soft unlithified sediments. The V_s variation in the first tens of meters can be one order of magnitude greater than the V_p variation (Hamilton 1976). The lower velocity makes the wavelength shorter, providing a high resolution for body wave techniques.

However, the offshore acquisition of shear waves for refraction or reflection applications is challenging. The use of horizontal receivers and a horizontally polarized source presents difficulties even in very shallow water. Because the water does not propagate shear waves, sources and receivers have to be properly coupled to the seafloor. Deploying receivers at the bottom is a standard practice in marine and seabed exploration applications, with hydrophones and geophones either in cables (ocean bottom cable) or in independent nodes or stations (ocean bottom station). Converted waves from compression sources in the water can be used for the shear imaging and characterization. An alternative is the use of specific bottom dragged instrumentations able to generate and detect shear waves (Stoll and Bautista 1994; Winsborrow et al. 2003) and Love waves at the seafloor.

The analysis of Scholte wave propagation is an effective alternative for the estimation of the shear wave velocity of the shallow subseafloor. Scholte waves are the surface waves propagating at the boundary between a solid and a liquid layer. They are similar to Rayleigh waves and can be analyzed for the underwater characterization with the same methods used for

Rayleigh waves. The shear wave velocity profile can be inferred inverting the dispersion relation obtained by processing the pressure wave field associated with interface waves. The generation and recording of Scholte waves in seabed applications can be performed with a simple experimental set-up, with no need for bottom sources.

8.2.1 Scholte waves

A system consisting of a liquid layer over a solid layered half-space allows two types of interface waves to be propagated: Scholte waves, the properties of which mainly depend on the shear wave velocity of the sub-bottom sediments, and P-guided waves, the properties of which mainly depend on the compression wave velocity of the sub-bottom sediments and of the water layer. Both these events are excited by sources in the water or at the seafloor and can be recorded as a pressure or particle velocity wave field at the seafloor. Schematically, the two events are represented in Figure 8.9, in which the qualitative shape of the displacement eigenfunctions for the fundamental mode of propagation is reported.

The Scholte waves (Scholte 1947) are the normal modes, that is, the eigenvalues related to real roots of the dispersion equation in the *soft bottom* case, when the shear velocity of the solid is lower than the compressional velocity in the liquid (Carcione and Helle 2004). They propagate parallel to the water–sediment interface, with most of the energy traveling at the solid–liquid interface. They are dispersive and modal and are similar to Rayleigh waves at the solid–air interface and to Stoneley waves at the solid–solid interface (Ewing et al. 1957).

The guided waves correspond to the leaky modes, with energy partially trapped in the waveguide. The SV reflections are supercritical, while the *P* is subcritical, and a fraction of the compression energy is radiated down out of the low-velocity layer. In most shallow sediments, with a high Poisson ratio, these waves degenerate toward the simple modes of acoustic guided waves. As Scholte waves, they are modal and dispersive, but even the fundamental

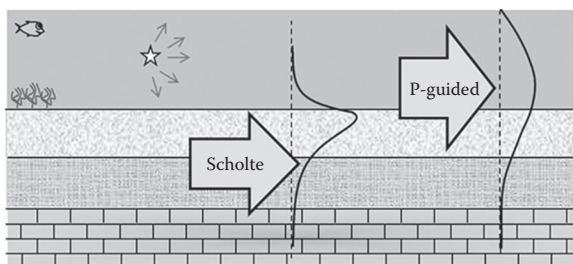


Figure 8.9 Schematic representation of interface waves propagating at the seafloor.

mode has a cutoff frequency that can be quite high compared to the usual frequency ranges of conventional seismic prospecting. The phase velocity is higher than the V_p of the water at any frequency, and it depends on the compressional wave velocity of the sediments.

The particle motion of Scholte waves is similar to that of Rayleigh waves. The energy is concentrated at the interface and decays rapidly with distance from the interface. For the case of two half-spaces in contact, the penetration is roughly equal to one wavelength in the solid and to a half wavelength in the liquid. In Figure 8.10a, the normalized displacement is plotted versus the normalized depth (fraction of the wavelength). The particle motion is elliptical retrograde, such as for the Rayleigh wave, and it changes direction at a depth of about a tenth of the wavelength (Figure 8.10a). High Scholte wave amplitudes can be expected at the seafloor, where both the pressure field and the vertical component of the particle motion have their maxima.

One of the differences with Rayleigh waves is that Scholte waves are dispersive even with a homogeneous solid half-space. Moreover, given a sediment column, the Scholte wave velocity is lower than the corresponding Rayleigh velocity (i.e., in the same model without a water layer on top). In the low-frequency range, the influence of the water layer is reduced because of the long wavelength; hence, Scholte wave velocity tends toward the Rayleigh wave velocity in the solid sub-bottom sediments. In the high-frequency range, Scholte wave velocity tends toward the Stoneley wave velocity at the solid-liquid interface. The dispersion curve of the Scholte wave for different water depths is plotted in Figure 8.10b. The velocity mainly depends on the shear wave velocity of the sediments, as for the Rayleigh wave.

Scholte waves can be generated by a source at the interface, as well as by a pressure source in the water layer. A large air gun array can excite strong Scholte waves in 100 m of water. The excitability of the modes acts as a high-cut filter; lower and lower frequencies are propagated as the water layer becomes thicker and thicker. The larger the distance of the source from the seafloor, the lower the frequency that is excited. This aspect has to be taken into account to plan the survey with respect to the targets because short wavelengths are required for increasing the resolution close to the sea bottom and long wavelengths for deep characterization.

Generally, Scholte waves are observed as low-frequency and low-velocity events. Indeed, the shallow unconsolidated sediments can have a very low shear wave velocity, and consequently, very low velocity Scholte waves are observed. In Figure 8.11, the properties of the solution are shown for a simple three-layer model (two layers of thickness 3 m each and shear velocity of 200 and 300 m/s and a half-space with a shear velocity of 400 m/s; the water layer is 3 m thick; the P-wave velocities are 1500, 1550, 1600, and 1700 m/s, respectively). The Rayleigh wave dispersion curves for the same model with no water layer are also reported for comparison in Figure 8.11.

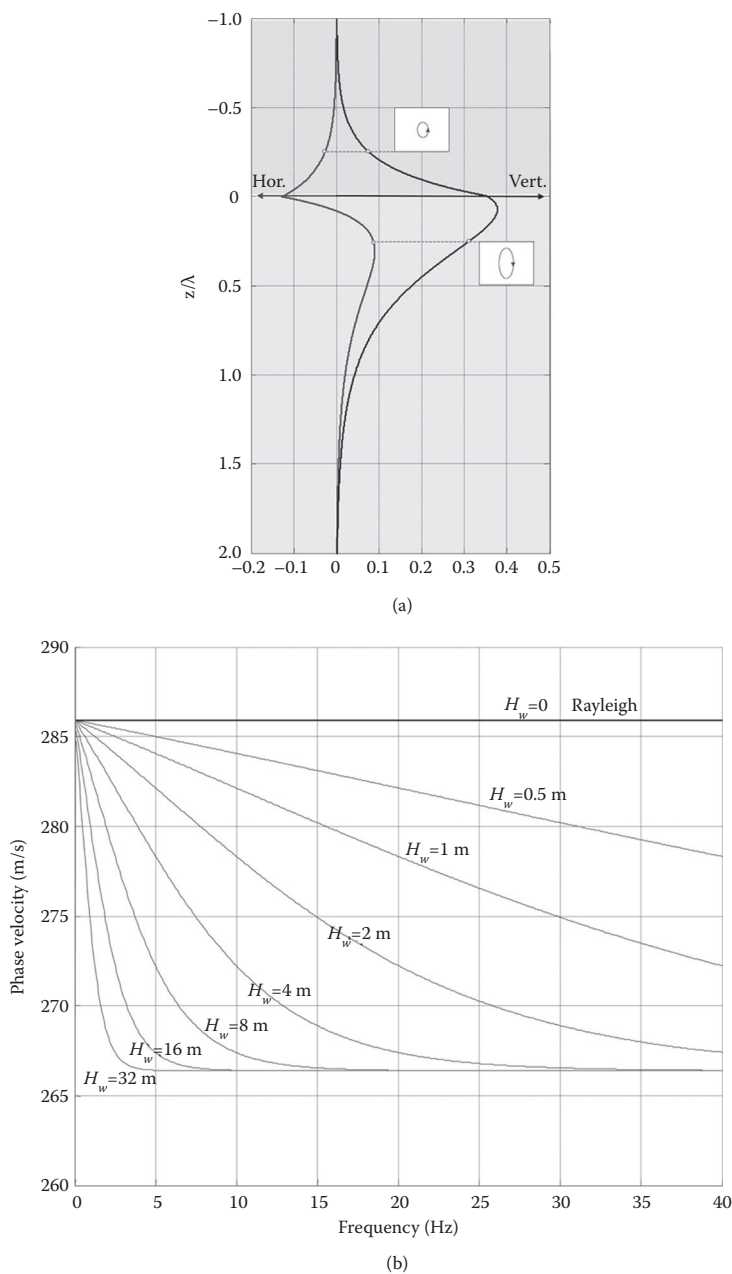


Figure 8.10 (a) Particle motion of the Scholte wave for the case of two homogeneous half-spaces in contact. (b) Dispersion curve for a bottom half-space ($V_s = 300$ m/s) with a top water layer of variable thickness.

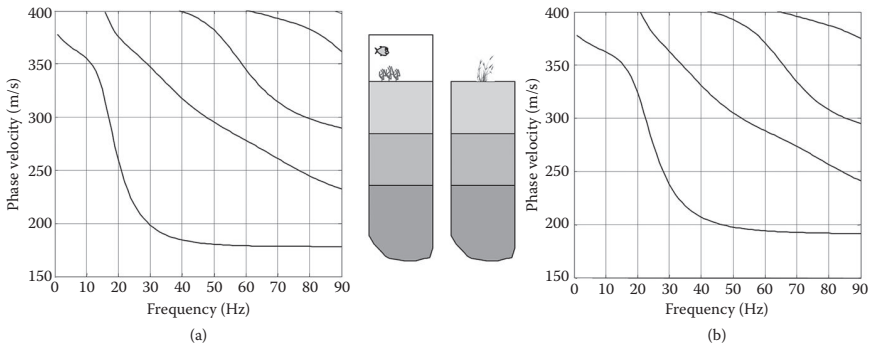


Figure 8.11 Example of (a) Scholte and (b) Rayleigh modes in a layered elastic medium with the same solid layers.

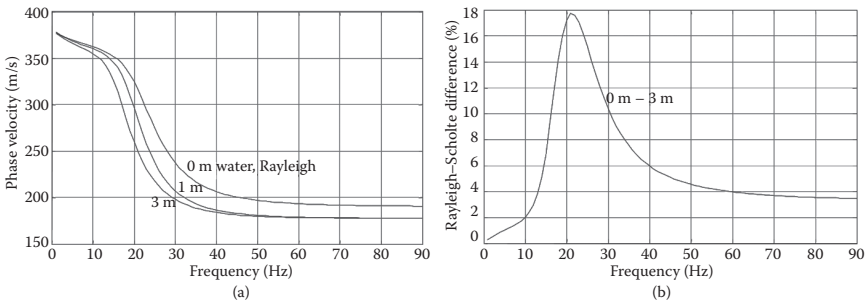


Figure 8.12 Example of Scholte and Rayleigh modes in a layered elastic medium with the same solid layers: (a) dispersion curves for different thicknesses of the water layer; (b) relative difference between Rayleigh and Scholte wave phase velocities for a 3 m thick water layer.

The fundamental mode and three higher modes exist in the considered frequency range for both types of waves. The effect of the water layer is a slight decrease of the phase velocity, particularly at high frequency. The Scholte fundamental mode does not have a cutoff frequency, and at a low frequency, its phase velocity tends to resemble that of Rayleigh waves.

A parametric analysis is shown in Figure 8.12 where the dispersion curves of the first normal mode for a layered system with different thickness of the top water layer are reported. The zero thickness case represents the Rayleigh wave solution. Even a rather shallow water layer affects the phase velocity. The analysis of marine data with a Rayleigh wave forward modeling is not correct, and it may lead to underestimation of the sediment velocity. The difference in phase velocity is a function of the frequency, as shown in Figure 8.12b for a 3 m thick water layer. Moreover, the thickness

of the water layer may be variable along the survey line, especially in shallow waters. If this variability is not properly taken into account for the analysis, the results may be affected by artifact with false apparent variations of the sediment thicknesses.

A factor that is important to take into account in the Scholte wave inversion is the P-wave velocity of the sediments. In the inversion, if the Poisson's ratio is assumed a priori, its value should be compatible with saturated materials (see also the discussion in Section 7.1) unless trapped gas is expected at the specific site.

8.2.2 Guided waves

The complex roots of the dispersion equation of the Scholte problem correspond to leaky modes of propagation, with a phase velocity higher than the highest shear velocity in the system. The real part of the complex wave-number corresponds to the acoustic propagation, and, under some hypotheses, the imaginary part can be neglected. It can be shown that, with a high Poisson ratio, the leaky modes can be approximated by the normal acoustic modes. The wave field is essentially composed of multireflected P-waves. It consists of compressional dispersive guided waves, the phase velocity of which mainly depends on the P-wave velocity of the layers and those that propagate with reduced attenuation. In the domain of ocean acoustics, they are sometimes called ultra low frequency (ULF, in the range 1–100 Hz). For shallow waters, the frequency range is shifted toward higher values. The acoustic guided waves are the total internal reflections in a liquid layer over a liquid half-space with higher velocity. The elastic half-space affects the dispersion with a low Poisson's ratio.

In the following example, a 20 m thick liquid layer with a velocity of 1500 m/s rests above a half-space with a P-wave velocity of 1700 m/s. The dispersion curves for the acoustic case (liquid half-space, i.e., $V_s = 0$ m/s) and for a solid elastic half-space (of variable V_s) are plotted in Figure 8.13. With a high Poisson's ratio (corresponding to $V_s = 300$ m/s), which is typical for soft uncemented sediments, the behavior is similar to that of the acoustic case.

The sub-bottom layering has to be considered when an accurate representation of the guided wave dispersion is required (e.g., if the experimental dispersion curves are inverted to obtain a velocity model). As an example, let us consider a simple model with 3 m of water ($V_p = 1500$ m/s, $V_s = 0$) and a 3 m layer of sediments ($V_p = 1560$ m/s, $V_s = 60$ m/s) above a bedrock ($V_p = 2000$ m/s, $V_s = 1000$ m/s). The acoustic approximation and the elastic solutions are plotted together in Figure 8.14. The acoustic approximation shows the general trend, the upper and lower velocity limits, but it fails to reproduce the shape of the curves. Considering the fundamental mode, it is evident that the frequency range needed to estimate the velocities of the two solid layers is wide (from 100 to 1000 Hz).

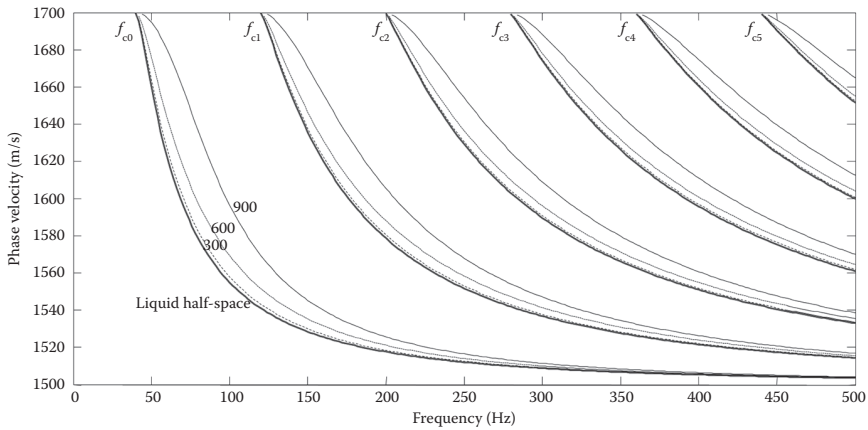


Figure 8.13 Dispersion modal curves for a 20 m thick water layer above a half-space. The acoustic approximation is valid for a high Poisson's ratio. For lower Poisson's ratios, or higher shear velocities in the half-space, differences cannot be neglected.

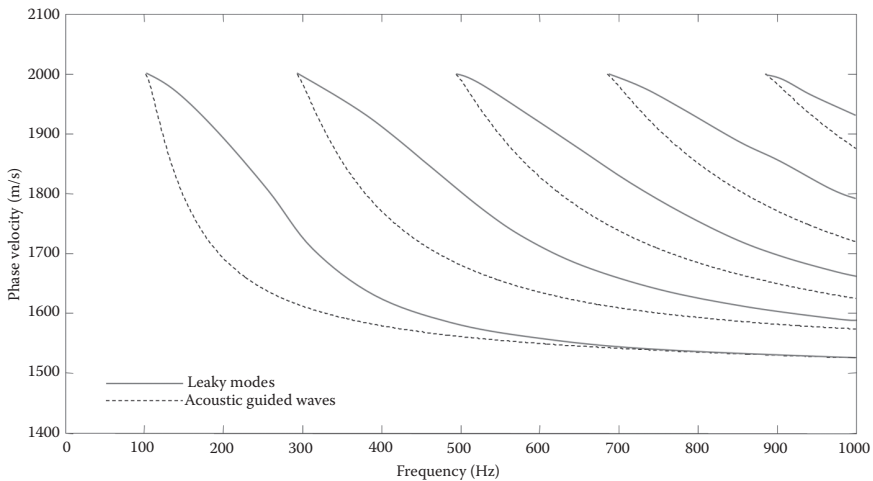


Figure 8.14 Modal curves for the guided P-waves for a two-layer system.

For wavelengths greater than the layer thickness, the fundamental mode is dominating, although higher modes become important for shorter wavelengths. It can be noted that the acoustic waves exist only above the cutoff frequency of the first mode. It can be computed with a simple relation for a single-layer waveguide. In this example, the cutoff frequency of the fundamental mode is about 100 Hz.

The P-wave velocity of the sub-bottom strongly affects the guided wave dispersion. The P-wave velocity of unconsolidated sediments can be close to the P-wave velocity of the water; hence, the a priori information about the water layer is crucial.

8.2.3 Example

The acquisition of data can be performed using strings of hydrophones at the seafloor or in towed streamers and different marine sources, including air guns, sparkers, boomers, and so on. The discussion of marine acquisition systems and technologies goes beyond the scope of this section.

In nearshore applications, the possibility of deploying the source on shore should be considered if the offset range makes it possible. Passive data can also be profitably used in nearshore and offshore configurations.

While designing the data acquisition, it is important to consider that Scholte waves and guided waves can be used in a joint inversion to estimate the shear and compressional wave velocities, better constraining the geometrical parameters. A joint acquisition requires a proper sampling, in time and space, of the two events. Indeed, the phenomena are characterized by well-separated ranges of frequency and velocity. Scholte waves propagate at a phase velocity close to the shear velocity of sediments, while guided waves propagate at a velocity higher than the compressional wave velocity in water. Moreover, the guided waves can have a cutoff frequency higher than the maximum observed Scholte wave frequency. The acquisition of the full wave field can be performed by recording the associated pressure field, using hydrophones at the bottom. The vertical component of the particle velocity can be used as well, but the coupling of the hydrophones is usually better, leading generally to more accurate data.

In the following, an example of acquisition is provided. A string of 12 hydrophones, spaced at 2 m, is deployed on the seafloor to detect the guided waves generated by a simple impulse source in water—a speargun shot at a metal plate. The central frequency of the acquired traces is higher than 500 Hz, and the S/N is high over a wide frequency band. The wavelength of the propagating waves ranges from less than 1 to 1.5 m, creating spatial aliasing. The dispersion curves extracted from the analysis of the fast event, in the first 25 ms, are shown in Figure 8.15.

The shot gather is dominated by the high-frequency components related to guided waves, with a phase velocity greater than 1500 m/s, but it also contains Scholte waves. This event, propagating with a velocity of less than 50 m/s, is visible in the first traces and can be enhanced by a simple low-pass filtering. The first two traces are shown in Figure 8.16.

The presence of the two phenomena is much more evident in a raw record gathered across the shoreline (Figure 8.17). The first 12 traces are gathered

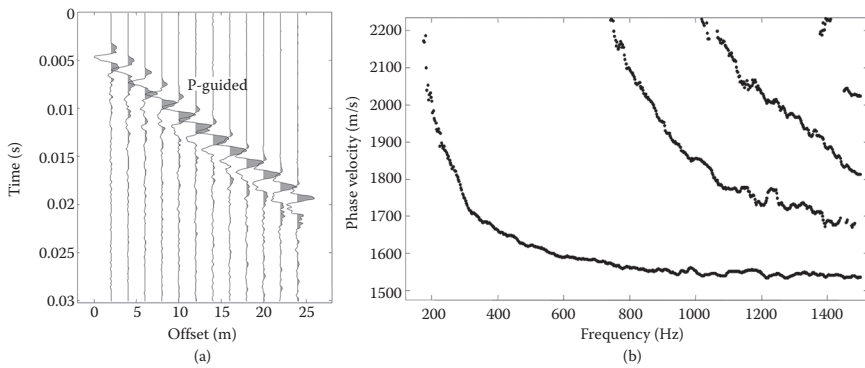


Figure 8.15 Example of marine data: (a) shot gather; (b) experimental dispersion curves for guided waves.

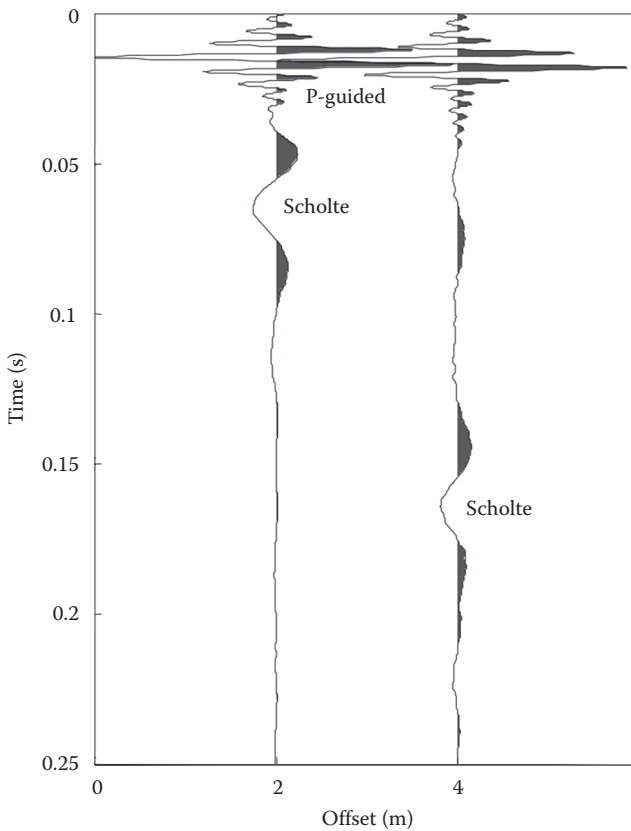


Figure 8.16 The first two traces of the shot gather of Figure 8.15a after low-pass filtering.

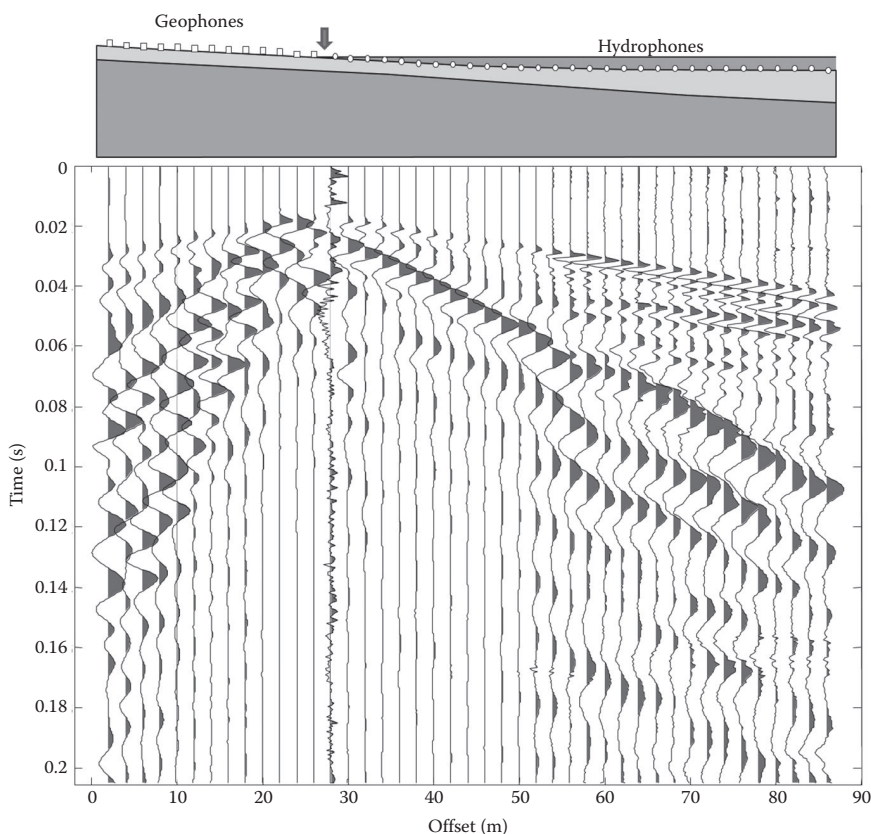


Figure 8.17 Example of experimental data collected along a shoreline (the arrow represents the position of the seismic source).

by 12 geophones on land; the other traces correspond to hydrophones at the sea bottom. The source (a sledgehammer) is located on the strand. Even if the thickness of the water layer is gradually varying and some lateral effects (e.g., reflections) are present, the shot gather clearly identifies the presence of Rayleigh waves on land and of Scholte waves on the seafloor. Both Scholte and guided waves have been recorded in the marine portion of the array, and this is also confirmed by visual analysis of the time-domain traces.

If guided and Scholte waves are acquired, a joint inversion can be performed. The coupling of the two models via the geometrical parameters (thickness of the layers) is easily implemented. The general aspects of joint inversions are discussed in the next section.

8.3 JOINT INVERSION WITH OTHER GEOPHYSICAL DATA

8.3.1 Joint inversion

The integration of multiple datasets and measurements is a crucial step in geophysical exploration. Multiple physical properties are estimated and used to quantify petrophysical parameters and for the discrimination of lithologies. Often even the imaging and identification of some targets require multiphysical surveying approaches. Gravity, magnetic, electromagnetic, and seismic methods are commonly used together in oil and gas exploration. Moreover, multiple datasets are often extracted from the same data acquisition or measurement; an example is the extraction and analysis of reflected and refracted body waves and surface waves from the same seismic dataset (see Section 7.5).

The integration and homogenization of multiphysics data into a consistent model can also improve the resolution, joining methods with different sensitivities to different subsurface features and to different portions of the investigated domain. Moreover, it can mitigate the nonuniqueness in the single-domain inversion of the individual measurements. With multiple physical domains, including a direct or indirect relationship between the parameters, it is possible to reduce the multidimensional model space to a smaller subspace.

The simplest integration of multiple datasets is based on the superposition and comparison of individual models for cross-validation. Still, the reconciliation of multiple models may be challenging and ambiguous. The synergies between methods and datasets can be exploited using mutual constraints, extracting from each the most reliable and trusted features. For example, a structural geometrical model can be built from one measurement and imposed on the other single-domain inversion. These approaches are subjective and rely on the operators' decisions, and hence several subjective solutions are possible. A more rigorous and robust approach for the integration of multiple measurements is based on joint inversions of the different datasets.

The term joint inversion indicates the process of estimating the unknown model parameters from multiple sets of geophysical data, building a unique objective function. The multiple physical domains of an Earth model are jointly estimated fitting multiple datasets. A *single-domain joint inversion* uses multiple datasets to estimate a model with a single physical domain or a type of unknown. For example, a single-domain joint inversion can be used to estimate a V_S model from the measured propagation properties of Rayleigh and Love waves. A *multiple-domain joint inversion* aims at the estimation of an Earth model with multiple properties; for example, to estimate a unique model with V_P and V_S from Rayleigh waves and P-wave refractions.

Different approaches to the joint inversion have been proposed. The *cooperative joint inversion* (Lines et al. 1988) essentially iterates inversion steps between the two domains. The result of one inversion step in one domain is converted and used as an initial model for the other domain. The process is repeated until convergence.

The *simultaneous joint inversion* inverts the datasets jointly with a link between the model domains. The links can be essentially of two kinds: geometrical or structural on one hand, intrinsic or petrophysical on the other hand.

The geometrical simultaneous joint inversion links the domains in terms of similarities among the geometrical distributions of model parameters. This can be achieved by minimizing the differences among the property distributions (for example, computed on the Laplacian of the model fields) (Haber and Oldenburg 1997). The cross-gradient links are also used to impose structural similarity across domains (Gallardo and Meju 2004).

The petrophysical link assumes a local intrinsic function among the domains. Analytical or empirical relationships may be used to link the physical parameters. These links may be either used as a priori constraints with fixed values or may be included in a global misfit function. In the latter case, the parameters of the petrophysical relationship become additional unknowns in the solution of the inversion problem (Jegen et al. 2009; De Stefano et al. 2011).

A schematic representation of the concept of joint inversion is provided in Figure 8.18. In Figure 8.18a, two datasets are inverted separately, and the two models are made compatible considering a possible link between them and the overlap between the two models. In a simultaneous joint inversion approach, the link is an input, together with the

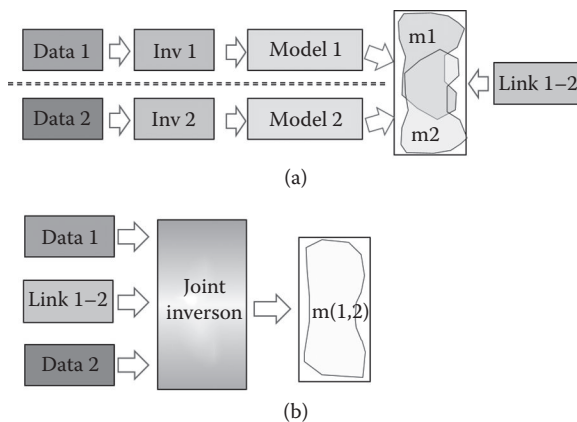


Figure 8.18 Schematic representation of (a) single-domain inversion and (b) joint inversion.

two experimental datasets, for the solution of the inversion process that produces a single model.

In the following section, we discuss different approaches to the joint inversion with examples of surface wave inversion with other geophysical data.

8.3.1.1 Geometrical joint inversion

The simplest link between the domains is given by the geometry of the models. Indeed, the spatial distribution of geophysical parameters is often controlled by common geological features. Parameters that appear as apparently uncorrelated can show similar spatial structure.

The geometrical resemblance can be included in the joint inversion, introducing structural links that encourage the similarity between the two models, without imposing or assuming any local relationship between the parameters. The resemblance between the two models can be defined, for example, using the cross-gradient approach (Gallardo and Meju 2003). The cross-gradient function is defined as the cross product of the gradients of the physical properties m_1 and m_2 in the two domains

$$\bar{\tau} = m_1(x, y, z) \times m_2(x, y, z) \quad (8.1)$$

The cross-gradient function measures the similarity between two images. It is minimum with the collinearity between the two domains. Figure 8.19 shows an example of two fields and the corresponding cross-gradients. The two fields m_1 and m_2 are plotted in Figure 8.19a and b, and the cross-gradient is shown in Figure 8.19c. The cross-gradient has low values in the portion where the dominant direction is similar (bottom part), even if the values are not directly positively correlated, and it has high values where the two fields show different geometrical distributions.

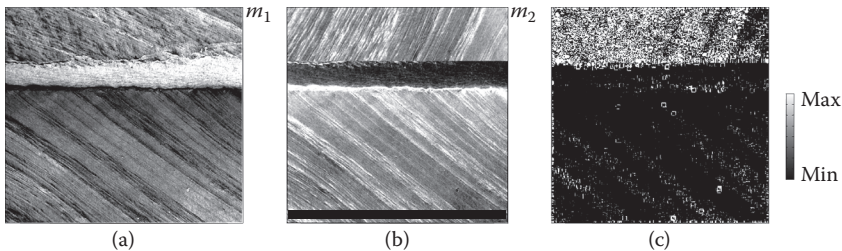


Figure 8.19 The similarity of the spatial distribution of two physical parameters can be evaluated using the cross-gradient: (a) model m_1 ; (b) model m_2 ; (c) cross-gradient $m_1 - m_2$.

The joint inversion of the two domains can be performed, defining an objective function based on the optimization of the two datasets and of the geometrical similarity (for example, the residuals of the different datasets and the cross-gradient field). The structural link can be part of the objective function (Tryggvason and Linde 2006), or it can be imposed as an equality constraint, limiting the model search in the iterative process to the subspace of models with geometrical similarity.

In a 1D layered model, the geometrical joint inversion implies similar layer thicknesses, regardless of the fact that the geophysical parameters of each layer can be completely uncorrelated.

8.3.1.2 Petrophysical joint inversion

When the different domain parameters have an intrinsic petrophysical link, the domains can be linked using a local parameter relationship. In the petrophysical joint inversion, a local relationship among the multiple-domain model parameters is used. These links can be model-based physical laws or empirical relationships among geophysical parameters. Widely used relationships include Archie (1942), Wyllie et al. (1956), and Gardner et al. (1974).

In the petrophysical joint inversion, the objective function can be built from three different elements: the residuals of the different domains, the single-domain constraint and regularizations, and the cross-domain links. The petrophysical link can be imposed as an equality constraint or encouraged minimizing the local residual between an a-priori relationship and the model relationship. The three elements of the objective function can be combined using different weights. The different domains can be preconditioned and regularized independently, and multiple scales can be solved simultaneously.

8.3.2 Surface wave joint inversion

Surface wave methods rely on an inversion stage to estimate the unknown model parameters (i.e., shear wave velocity and damping ratio). The coupled inversion of velocity and attenuation described in Chapter 6 can be considered a joint inversion. In this paragraph, we discuss the joint inversion of surface wave data with other geophysical measurements, seismic and nonseismic. The joint inversion can be used to integrate the surface wave data with other geophysical data. This approach improves the estimation of soil properties to which the surface wave measurement has limited sensitivity. Moreover, the inclusion of additional information allows solution nonuniqueness of the inverse problem to be mitigated.

In the following section, joint inversions with electrical, electromagnetic, and seismic data are discussed. Other joint inversions have been proposed in the literature—for example, with microgravity data (Hayashi et al. 2005).

8.3.2.1 Joint inversion with electrical and electromagnetic measurements

The joint inversion of the surface wave dispersion with 1D electrical and electromagnetic data can be used with, for example, geometrical links to impose common boundaries. Lithological boundaries in the shallow subsurface affect the shear wave velocity and the resistivity of shallow layers, and a geometrically coupled inversion is possible (Hering et al. 1995). The measured data have similar properties. The apparent resistivity curves from vertical electric sounding (VES) and time-domain electromagnetic (TDEM), similar to the dispersion curves, provide integral information about the corresponding physical parameters. Moreover, both forward models are based on a global transfer matrix built from layer matrices for the 1D solution.

With reference to the integration of VES and surface waves, the two sets of measured data can be the apparent resistivity versus the electrode array geometry and the Rayleigh wave phase velocity as a function of frequency. The model parameters can be the thickness, shear velocity, and resistivity of each layer.

The choice of geometrical coupling implies that interfaces (layer boundaries) are similar or coincident. However, this does not force to have a variation of one parameter when the other varies. For instance, the water table is a major interface for resistivity, but it is not for the shear wave velocity.

The model parameters can be joined in a single vector, creating a single Jacobian matrix with the partial derivatives for both set of parameters. The rows correspond to the p values of apparent resistivity ρ_{app} plus the m values of phase velocity v_r . The columns contain the derivatives with respect to the n resistivity values ρ , the $n-1$ thickness values h and the n shear wave velocity values V_s (for a model of n layers including the half-space). The absence of a petrophysical link leads to the zero values of the cross-domain derivatives.

An example is reported in Figure 8.20, in which a comparison between results from independent and joint inversions is shown. In this case, the tests were aimed at the identification of local stratigraphy in a geological context with alternate layers of sandy and clayey layers. In particular, the expected thickness of a shallow clayey layer was of interest for the assessment of the risk of seepage below a river embankment (Comina et al. 2004). The clayey layer is identified on the basis of its lower resistivity at a depth between 8 and 13 m. The solution of the joint inversion is deemed to

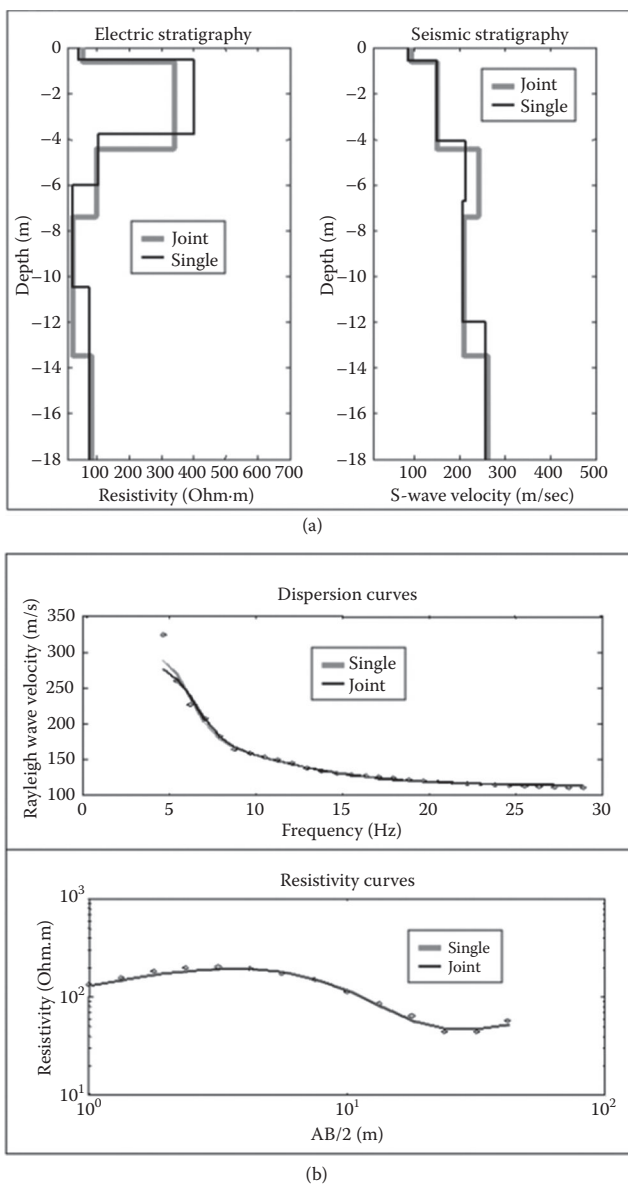


Figure 8.20 Joint inversion of surface wave and electrical resistivity data: (a) electrical and seismic profiles from joint and individual inversions; (b) comparison between experimental and numerical dispersion and resistivity data for joint and individual inversions. (From Comina, C. et al., *Proceedings of ISC-2 on Geotechnical and Geophysical Site Characterization*, Millpress, Rotterdam, pp. 451–458, 2004.)

be more robust and more reliable because it honors simultaneously the two experimental datasets.

A similar approach can be used for time-domain electromagnetic (TDEM) data and for magnetotelluric data with long-period surface waves.

8.3.2.2 Joint inversion with other seismic data

Considering seismic data, the joint inversion of Love and Rayleigh waves can be considered a single-domain joint inversion. The same layer model is constrained by two different datasets, and the unknowns can be the layer thickness and the shear wave velocity, under the assumption of isotropic behavior.

Particularly interesting is the case of the joint inversion of Rayleigh and guided waves. A joint inversion can be obtained with a global misfit function, where the Rayleigh and guided wave dispersion curves are used together. The distance between the picked curves and the determinant of the Haskell–Thomson matrix can be used, as discussed in Section 7.4. Alternatives include the use of the distance between curves modeled independently from the same stack of layers. An example is provided by Roth et al. (1998).

The joint inversion of surface waves and refracted P-waves is particularly interesting for several reasons: the two corresponding dataset are often available together, the two methods offer many synergies, and the intrinsic link between the two domains is often rather strong and dependent upon features of interest.

Actually, the two datasets can be extracted from the same seismic data, so the measurements can be performed simultaneously. They are normally acquired together in exploration seismic, and they can be acquired together in near-surface engineering testing, without significant additional operational effort. Because of the many basic differences between the two components of the wave field that are analyzed, the two methods complement each other in many respects (Foti et al. 2003).

Surface wave methods overcome some inherent limitations of the refraction technique caused by refraction equivalence such as the ones due to hidden layers, gradual velocity variations, and velocity inversions (Reynolds 1997). Also, the sensitivity to the shear wave velocity is particularly important in saturated sediments where the P-wave velocity is mainly controlled by the fluid compressibility.

On the contrary, refraction results provide very useful information for surface wave interpretation: constraints for surface wave inversion (e.g., the bedrock position), indication of the water table depth (which allows a good estimation of Poisson's ratio in surface wave modeling), and a warning in the case of lateral variation of velocities (e.g., topography of refractors).

When the two techniques give independently reliable results that are mutually confirmed, the information about compression and shear wave velocities allows the estimation of mechanical properties and porosity of the soil deposit. At the same time, the relationship between the elastic parameters can be used as a petrophysical link, or the similarity between the two spatial distributions can be used as a geometrical link.

8.3.2.3 Joint inversion of refracted and surface wave

Even if some benefits can be obtained just with the simple combined interpretation of the two results, the joint inversion provides a robust framework to exploit the synergies between the two methods.

In this section, we present an example of 1D joint inversion, where the refraction data suffer from equivalence due to the presence of a hidden layer. The seismic survey was carried out to determine the soil parameters for seismic site response assessment. The expected stratigraphy of the test site consists of a soft soil over a dense gravel layer and the presence of hard sandstone bedrock at a depth of less than 10 m.

The surface wave data show large dispersion and allow the extraction of a broadband dispersion curve, which is inverted for a layered model—confirming the presence of three main geological layers. The seismogram acquired with an unevenly spaced array, the extracted dispersion curve, and the inverted velocity profile are represented in Figure 8.21.

Refraction data acquired at the same location show a clear refraction in the first arrivals, with a large variation of velocity with offset. The picked travel times essentially show two velocities, with a large difference in velocity in the forward and backward shots, indicating the presence of a dipping layer.

The raw seismograms for the two end-off shots are shown in Figure 8.22. The interpretation of the travel time curves (Figure 8.22b) suggested a two-layer model, which is shown in Figure 8.22c. This result would locate a slightly dipping bedrock, of fairly high velocity, at a depth varying between 3.2 and 4 m, but the high velocity contrast (330 vs. 2600 m/s) between the two layers allows the presence of a hidden layer of significant thickness.

The “hidden layer” problem is one of the possible ambiguities of refraction methods. An intermediate layer might not be identified because its refracted arrivals reach the surface after the critically refracted waves from a deeper faster layer. In practice, the equivalence is due to the fact that multiple velocity profiles can generate exactly the same travel time curves. The example of Figure 8.23 shows how two different velocity profiles can generate exactly the same travel time curves.

The large difference in the bedrock depth (Figures 8.21 and 8.22) raises a further warning about the risk of a hidden layer. The evidence of dipping bedrock in forward and backward shots poses some questions

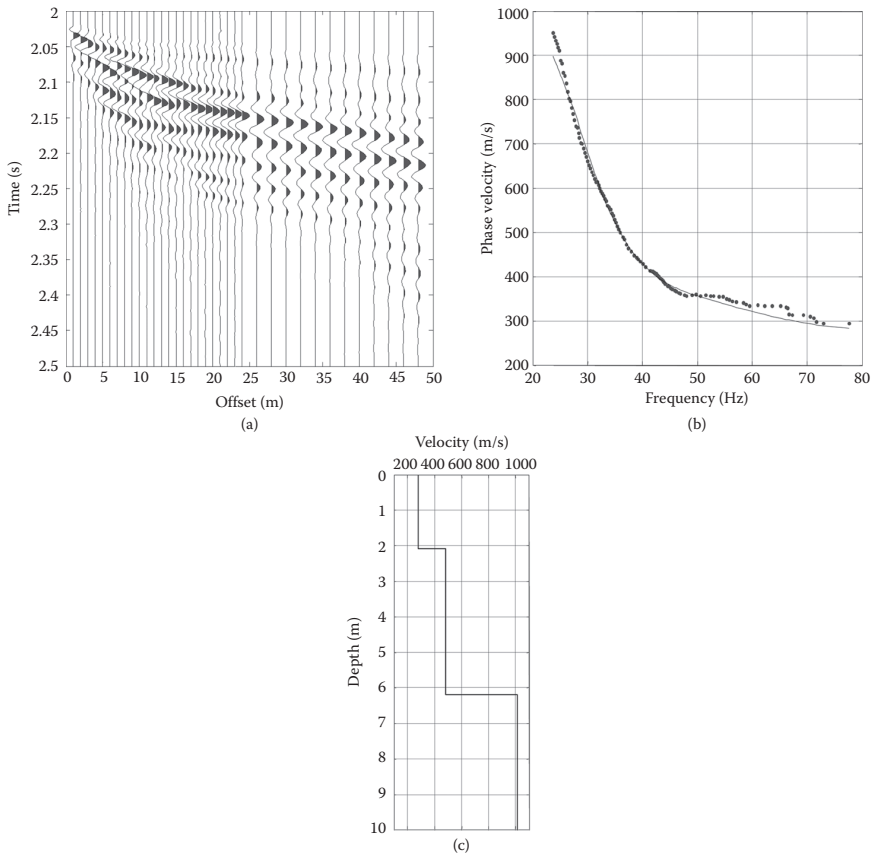


Figure 8.21 Surface wave data: (a) seismogram; (b) dispersion curves; (c) estimated shear wave velocity profile.

on the acceptability of the 1D hypothesis in surface wave interpretation. Essentially, the joint interpretation of the two measurements leads to a difficult reconciliation of the velocity profiles.

The two datasets (the picked travel times and the dispersion curve) are then inverted together, imposing common layer thicknesses for V_P and V_S models. A small dip is included among the model parameters to fit the travel time curves. The inversion is performed using a global Monte Carlo optimization, defining a joint misfit function as the weighted average of the two least-square misfits. The results are reported in Figure 8.24. The large contrast in shear wave velocity is confirmed, and the small dip is acceptable in terms of validity of the 1D assumption. The best 20 models are plotted in Figure 8.24c.

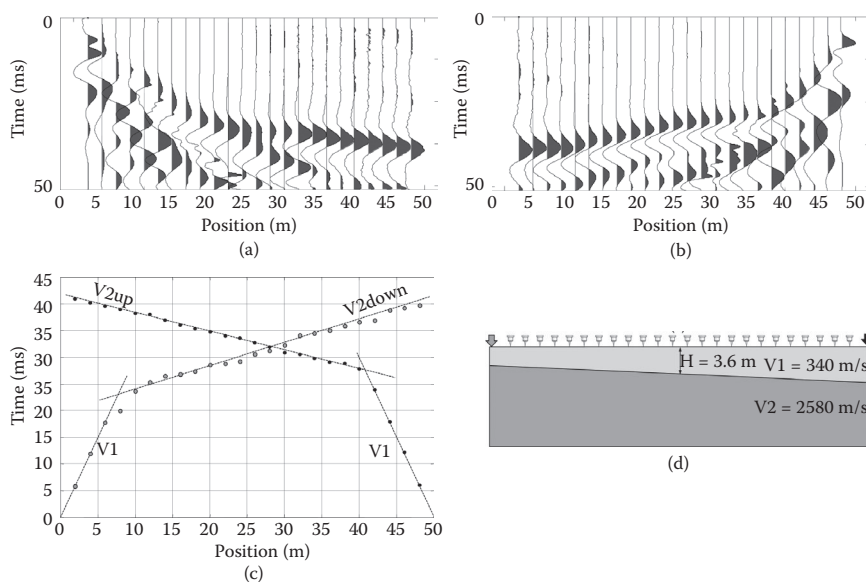


Figure 8.22 P-wave refraction: (a) seismogram for a forward shot; (b) seismogram for a reverse shot; (c) travel times and their interpretation with a two-layer model; (d) P-wave velocity model.

The joint inversion of surface wave data and P-wave arrival travel times can be implemented with conventional methods discussed in Chapter 6. The example reported in Figure 8.25 shows a comparison between individual inversions of the two datasets and joint inversion with a classical least-square approach (Piatti et al. 2013). The solution obtained with the joint inversion allows for an higher resolution and identifies the presence of a soft silt and clay layer at a depth around 20 m, which is embedded in between the top gravels and the bedrock below as confirmed by stratigraphic information obtained from borehole logs (Piatti et al. 2013).

8.3.2.3.1 Large-scale joint inversion of refracted and surface waves

In deep exploration, refracted body waves and surface waves are acquired over a large area using 3D geometries. The acquired traces can be processed to extract the two datasets—the surface wave propagation properties (typically, the dispersion volumes) and the refracted first arrival travel times. The two datasets have different spatial density, and the methods have different spatial resolution. The investigation depths can be different, resulting in two models with different resolutions and a partial overlap. A schematic representation of the sensitivity is provided in Figure 8.26. The travel times provide information on the P-wave velocity structure,

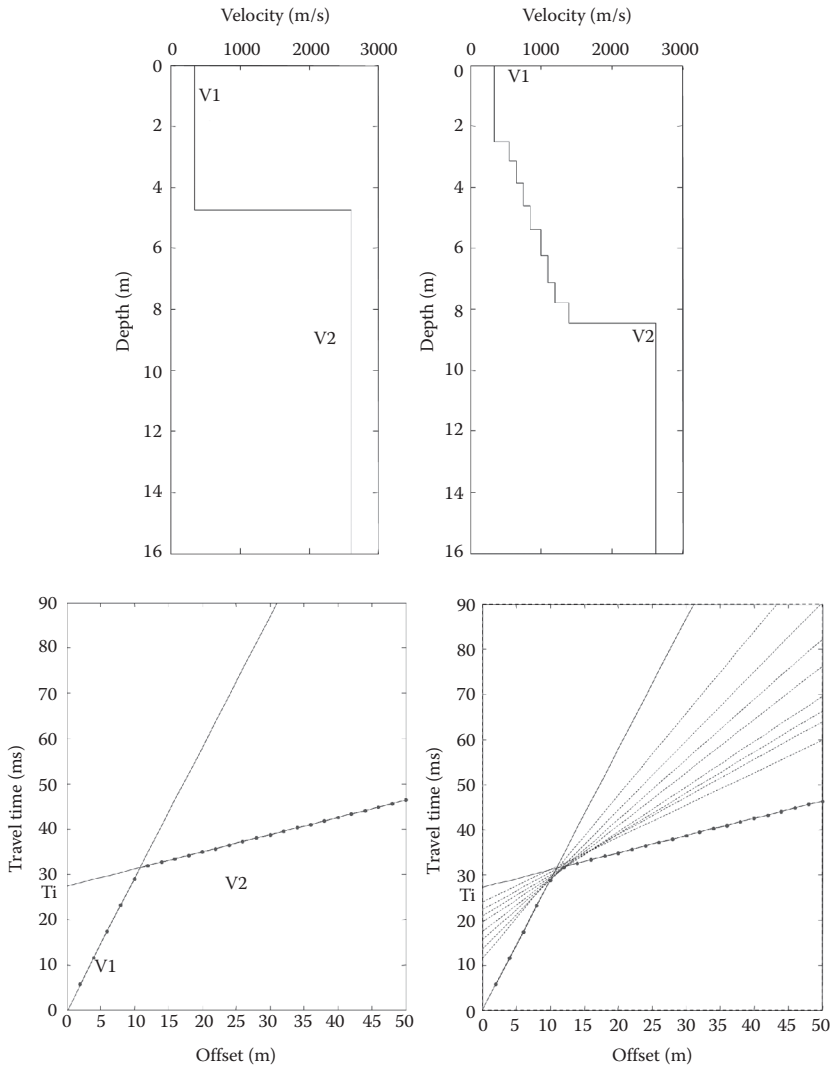


Figure 8.23 Equivalence of travel time curves for different profiles.

and the sensitivity is computed via ray-tracing in a 2D or 3D medium. The dispersion constrains the S-wave velocity structure, and the sensitivity is controlled by the eigenfunctions.

Both geometrical constraints and local intrinsic links can be used. A Poisson's ratio distribution can be used as petrophysical link. The example of Figure 8.27 shows a 2D application using a flexible joint inversion

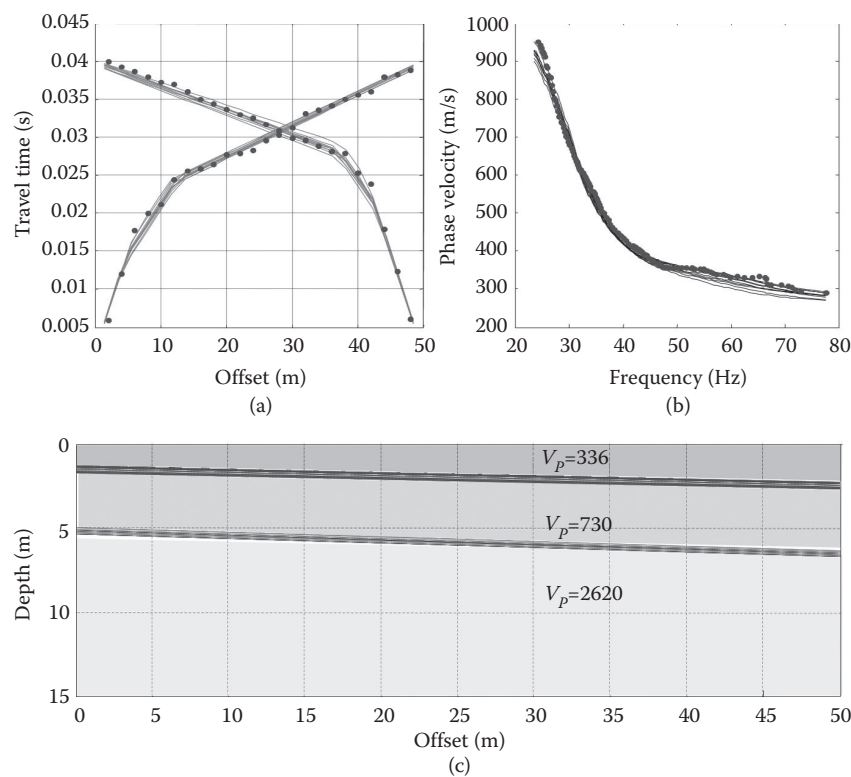


Figure 8.24 Monte Carlo inversion of the dispersion and the travel time curves: (a) fitting between theoretical and experimental P-wave travel times; (b) fitting between theoretical and experimental surface wave dispersion curves; (c) reconstructed seismic model.

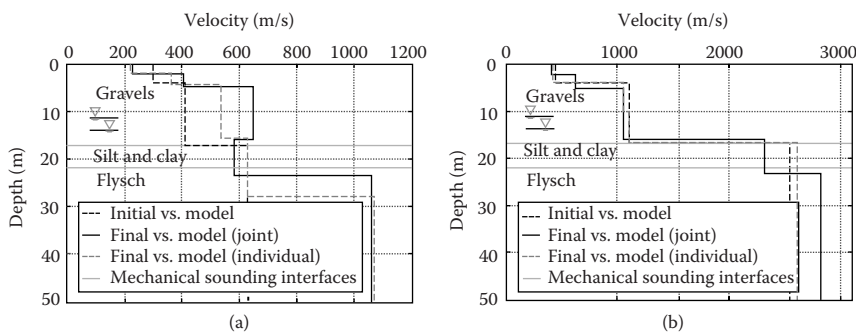


Figure 8.25 Joint inversion of P-wave seismic refraction and surface wave data: (a) S-wave model; (b) P-wave model.

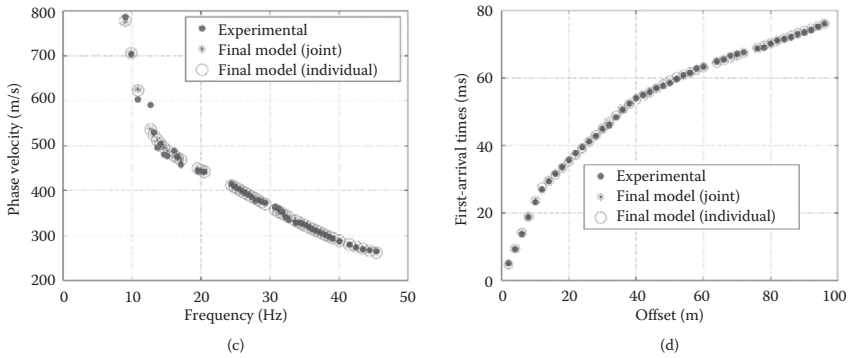


Figure 8.25 (Continued) Joint inversion of P-wave seismic refraction and surface wave data: (c) comparison between experimental and numerical dispersion curves; (d) comparison between experimental and numerical travel times. (From Piatti, C. et al., *Geophys Prospect*, 61(Suppl. 1), 77–93, 2013.)

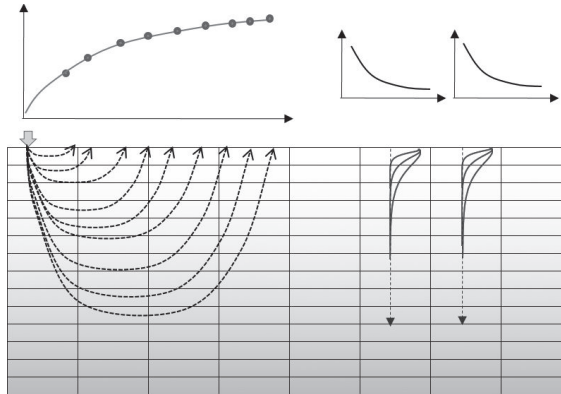


Figure 8.26 Schematic representation of the model sampling by refracted body waves and surface waves.

platform able to solve simultaneously for different scales and to handle separately the denser V_s unknowns to define the very shallow portions of the models and the coarser V_p unknowns to invert down to the main refractors of the model. The line, 10 km long, is located in Egypt along the Gulf of Suez and is discussed in Strobbia et al. (2010). The P-wave velocity model for the shallowest 400 m is plotted, together with a shallow stack section, in Figure 8.27. The correlation of the velocity distribution with the reflection stack section confirms the resolution of the obtained velocity distribution.

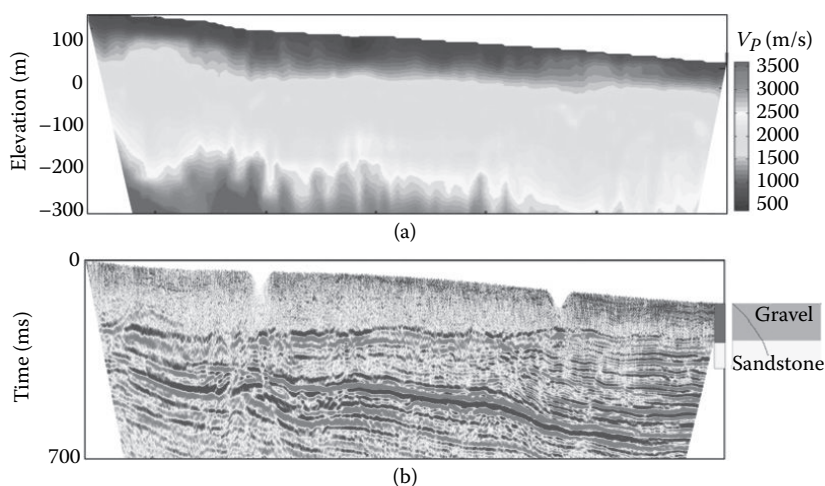


Figure 8.27 Joint inversion of surface wave and P-wave data: (a) velocity distribution from joint inversion; (b) stack section from P-wave reflection. (From Strobbia, C. et al., *First Break*, 28(8), 85–91, 2010.)

8.4 PASSIVE SEISMIC INTERFEROMETRY

Seismic interferometry has recently gained large popularity in many applications with both active and passive data. Generally, interferometry indicates the study of interference between pairs of signals. Seismic interferometry indicates the principle of generating seismic traces from virtual sources via cross-correlation or convolution of the observations at different receiver locations. The applications can have two main objectives: for the estimation or isolation of a component of the wave field and for the estimation of the properties of the medium in which the waves propagate. In this section, we discuss some aspects of passive seismic interferometry as a tool to analyze surface waves in ambient noise or microtremor data.

Recent theoretical studies have demonstrated that the diffuse wave field of the ambient noise can be processed and transformed, via cross-correlation, into a deterministic seismic impulse response, estimating the Green's function between receivers (Weaver and Lobkis 2001, 2002; Snieder 2004; Wapenaar 2004; Wapenaar and Fokkema 2006). Applications at the global seismology scale, with coda waves and long noise records, showed that the surface wave component of the Green's function between two receivers can be retrieved from the cross-correlation of observations at the receivers (Campillo and Paul 2003; Shapiro and Campillo 2004). The potential of this approach in the higher frequency range also has been proven at the engineering seismology scale. Application with active seismic data,

in the exploration seismology, confirmed the possibility of estimating the direct and the diffracted surface wave component of the wave field using interferometry (Halliday et al. 2010a).

The principle of seismic interferometry is rather simple. The summation of the cross-correlation between two receivers for multiple sources creates a signal that can be interpreted as the response that would be measured at a receiver location if a source was active at the other. The principle of interferometry is illustrated here with a simple example, following the review of Wapenaar et al. (2010a).

Let us consider two receivers, R_1 and R_2 , observing the surface wave signals from different sources S_i surrounding them. For each of the sources, the recorded signals $T(s_j, R_1)$ and $T(s_j, R_2)$ are cross-correlated, thus generating a trace $T_{12}(s_j)$. Such correlated trace depends on the position of the sources and on the properties of the subsurface around the receivers (Figure 8.28).

The process is repeated for a number of sources surrounding the two receivers, generating a multiplicity of correlated traces, one from each of the source points. These traces are collected in a correlogram gather (Figure 8.29); in the example, they are sorted according to the azimuth of the source. This gather shows the variation of the correlation according to the position of the source. The behavior is regular and smooth despite the difference in distance of the individual sources. Indeed, the correlation removes the effects of the propagation from the source to the receivers, and the correlogram gather is more regular than the two individual gathers of the traces at the two receivers because it mainly depends on the azimuth.

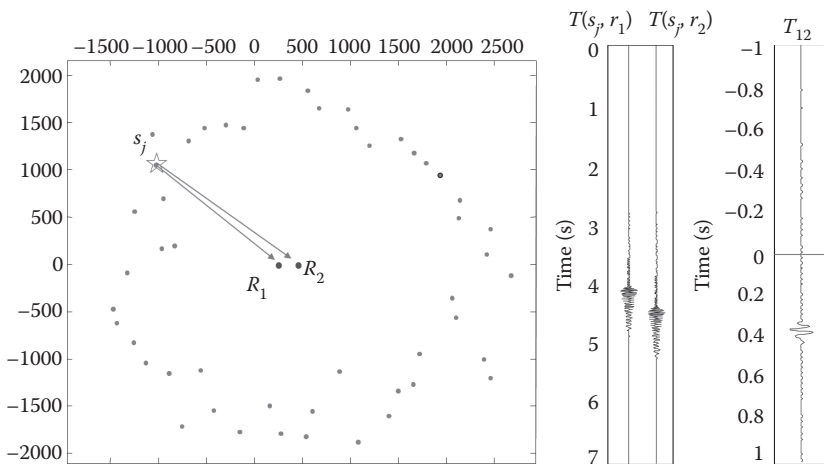


Figure 8.28 Two receivers surrounded by sources. For each source, the two traces are cross-correlated, creating a signal.

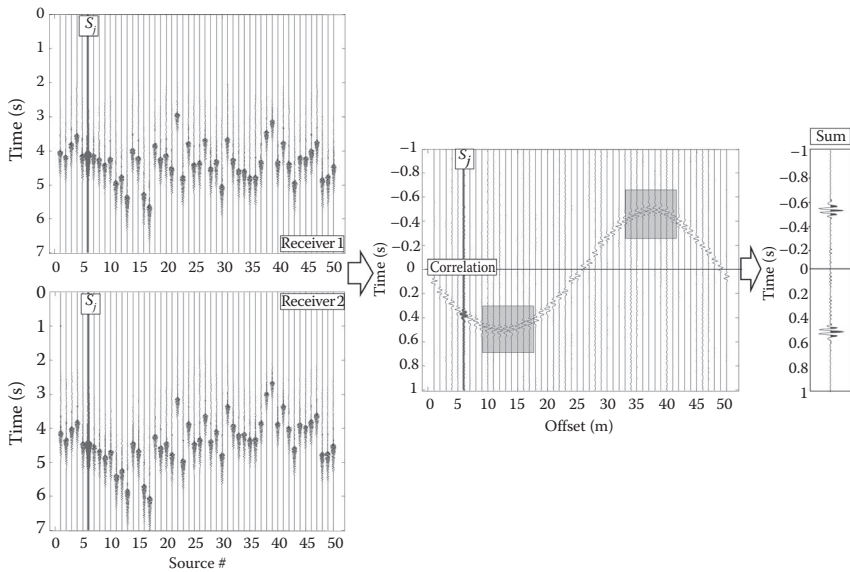


Figure 8.29 Seismic interferometry: signals received by the two detectors—their correlation and the sum.

In the summation of the correlated traces, contributions from opposite points tend to cancel out. The summation of the traces from the phase stationary points constructively produces a trace with the causal and the anti-causal response from the first receiver to the second receiver. In other words, the contributions from sources almost aligned with the two receivers do not cancel out. The resulting trace can be interpreted as the impulse response of a source placed at the first receiver, which is considered the virtual source.

The extracted signals can be interpreted and processed as virtual active data. One of the main applications of this process is the retrieval of the surface wave response from ambient noise and the subsequent analysis of the surface wave velocity (Shapiro and Campillo 2004). Most of the applications are based on the analysis of 2D sparse arrays and lead to a tomographic determination of the surface wave velocity distribution.

It has to be stressed that in case of a nonisotropic wave field, the use of a linear array does not allow correcting for the apparent velocity and can lead to largely overestimated velocities. But under the hypothesis of a diffuse ambient noise wave field, approaches similar to refraction microtremor (ReMi) can be used. The Green's function can be built from passive data with a linear array by using one detector as the virtual source. The advantage over ReMi approaches is that the seismogram can be processed as an active dataset. The validation of the hypotheses will be easier, and the corresponding f - k spectrum will have features more similar to the usual spectra from active data.

To compare the two approaches, simple synthetics are created using 1000 sources around a linear array of receivers in a layered medium. The interferogram is plotted in Figure 8.30. Its symmetry around $t = 0$ is due to the basic properties of the cross-correlation and confirms the uniform distribution of the sources.

The f - k spectrum of the interferogram is plotted in Figure 8.31. The maximum correctly indicate the position of the true wavenumber. The interferometric process cancels out opposite contributions, and the

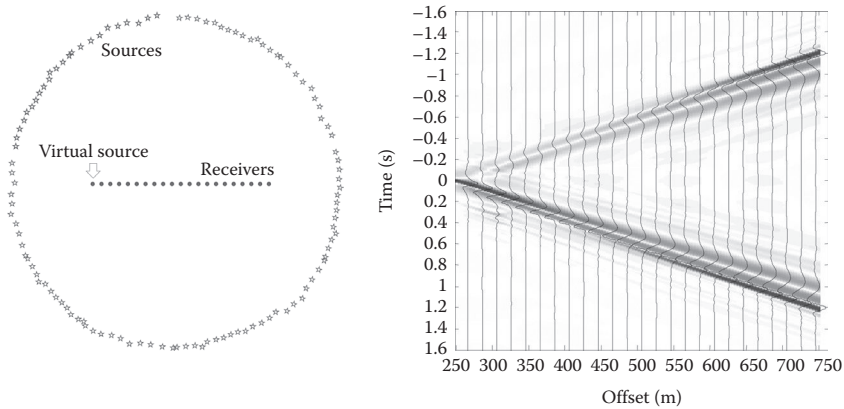


Figure 8.30 A linear array surrounded by sources, and the correlogram obtained from interferometric processing of 1000 sources.

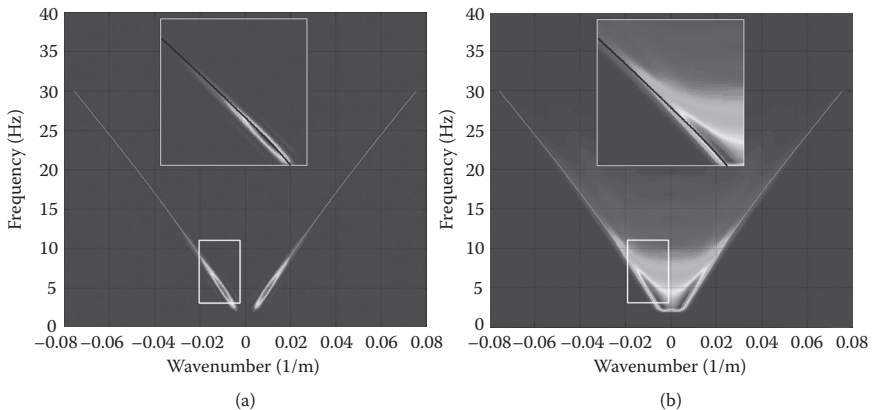


Figure 8.31 (a) The f - k spectrum of the correlogram gather of Figure 8.28. The maximum (black dots in the zoomed window) is located on the true theoretical wavenumber. (b) The ReMi spectrum obtained from the same synthetic data.

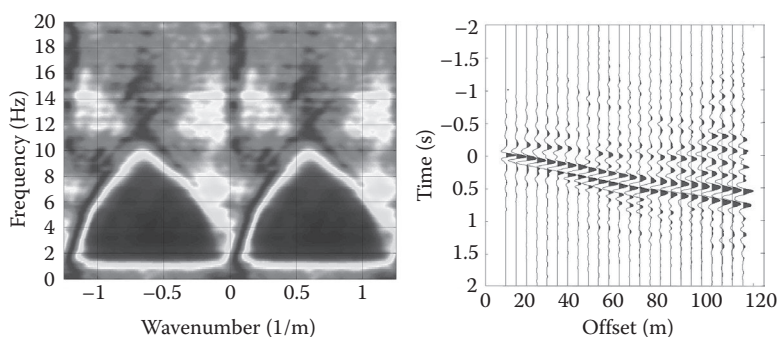


Figure 8.32 ReMi spectrum for a site where the ambient noise is dominated by a source in one direction and the correlogram.

spectrum has real maxima. As a comparison, the corresponding ReMi spectrum is plotted on Figure 8.31b.

The interferogram can help in validating the basic hypotheses of isotropic noise. For instance, in the case of a dominant source at a limited distance from the line, the obtained interferogram shows nonlinear events and raises a warning about the validity of results. An example of passive data dominated by sources in one direction is shown in Figure 8.32.

8.5 MULTICOMPONENT SURFACE WAVE ANALYSIS, POLARIZATION STUDIES, AND HORIZONTAL-TO-VERTICAL SPECTRAL RATIO

The amplitude of surface wave modal contributions as a function of frequency depends on the source location and spectrum, but it is also strongly affected by the site subsurface velocity structure. In principle, it is possible to use the measured amplitude spectrum as experimental data to estimate the site properties, but controlling or measuring the spectrum of the source radiation is not straightforward. However, the radial and the vertical components of the surface wave modal displacements have amplitude spectra that depend on the site properties. The ratio between the horizontal and the vertical components of the surface wave displacement only depends on the site properties for a given source location and direction. Therefore, the polarization as a function of the frequency can be used to estimate the site properties. In active surface wave testing, the spectral ratio or ellipticity of the Rayleigh wave can be estimated by analyzing the polarization of multicomponent active data. In passive data dominated by Rayleigh waves, the average ratio between the horizontal and the vertical components can be estimated over long records and used to evaluate the local site properties.

In a homogeneous medium, the ratio between the vertical and the horizontal components is constant, and it is a function of the Poisson's ratio. In layered media, the ellipticity is frequency dependent. The effect of the layering on the ellipticity is illustrated in Figure 8.33 (Boaga et al. 2013). A simple model, with a 25 m thick layer over half-space is considered; the velocity of the half-space is kept constant at 500 m/s, and the velocity of the layer varies from 500 m/s (thus giving a homogeneous half-space) to 200 m/s. The modal curves are plotted in Figure 8.33a, and the vertical ellipticity curves (V/H) for the fundamental mode are plotted in Figure 8.33b. The ratio between the vertical and horizontal components assumes a wider range for larger velocity contrasts; the polarization is more vertical in the high-frequency range and less in the low-frequency range. In the example, for a contrast larger than two, the horizontal component becomes larger than the vertical.

When the contrast is large, the amplitude spectrum of the vertical component has a strong peak, but it also has a zero (i.e., a frequency at which the vertical component vanishes and the motion is purely horizontal). The elliptical orbit changes direction across this point. Close to the maximum

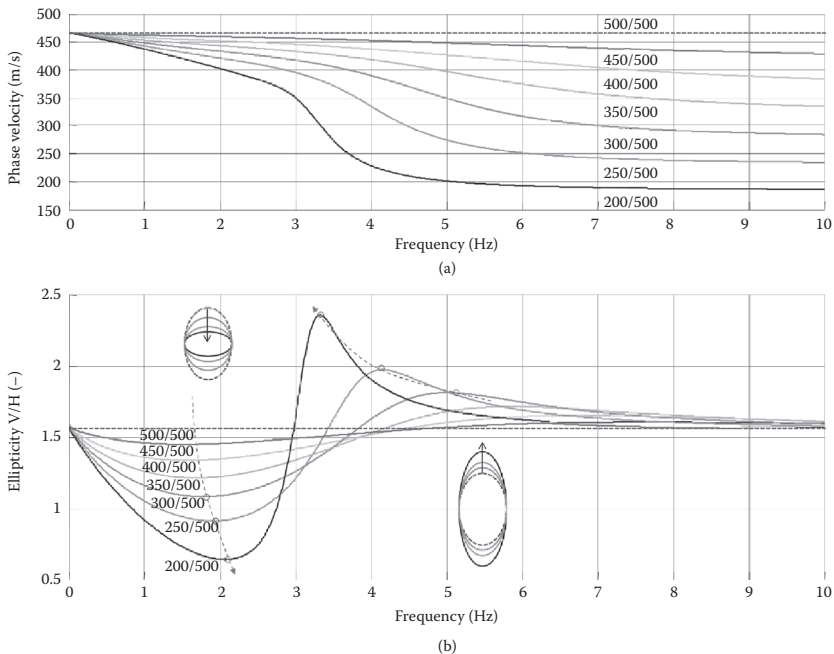


Figure 8.33 (a) Modal curves and (b) ellipticity of the fundamental mode for a single-layer model. The seven models have a 25 m thick layer with a V_s varying from 200 to 500 m/s, over a half-space with velocity 500 m/s.

of amplitude of the vertical component, the horizontal component vanishes and the motion is purely vertical.

Most applications use the horizontal ellipticity (i.e., the ratio between the horizontal and the vertical components, also called the H/V spectral ratio). The ratio between the horizontal and the vertical components for a simple model with a shear wave velocity ratio of four is plotted in a logarithmic scale in Figure 8.34: a low-velocity layer, 25 m thick, with 200 m/s shear wave velocity, over a 800 m/s half-space (V_p of 1500 m/s, 2000 m/s). In the very low-frequency range (A), the particle motion is elliptical and counterclockwise. At the first singularity, f_H , the motion becomes purely horizontal (B); in the intermediate frequency range, the motion is clockwise (C) and becomes more and more vertically polarized until the second singularity f_V where the motion is purely vertical (D) and the ratio H/V goes to zero. In the high-frequency range, the polarization tends toward the value of the first layer.

In the case of large impedance contrasts, the singularities of the spectral ratio are a strong signature of the site properties and have a very direct dependence on the thickness and velocity of the low-velocity sediments.

The peak of the spectral ratio H/V, where the vertical component vanishes, is important for different reasons. For one, the frequency of the H/V peak can be measured from passive multicomponent data. Second, the frequency of the peak can be related, for sites with a high impedance contrast, to the natural frequency of the site.

The natural frequency of the site is the fundamental shear wave resonance frequency. It refers to site transfer function for incident plane vertically propagating S-waves routinely used in 1D local site amplification assessment (Kramer 1996).

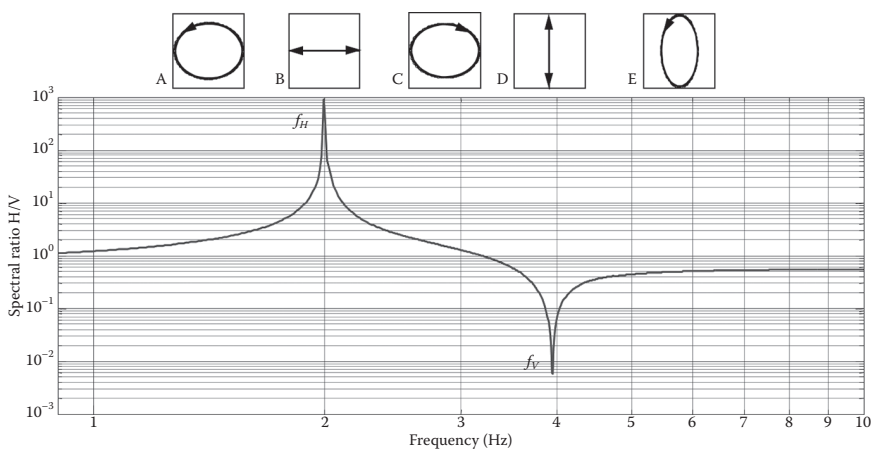


Figure 8.34 Horizontal ellipticity (H/V) of the fundamental mode for a simple model with 25 m at 200 m/s and a half-space at 800 m/s.

With a homogeneous low-velocity layer with a shear velocity V_s and a thickness H , the natural frequency is equal to the $V_s/4H$, and the peak of the ellipticity of Rayleigh waves tends to be very close to this value. In case of gradient or layering in the low-velocity layer, the zero of the vertical component is at a slightly higher frequency than the fundamental frequency. The Poisson's ratio and the mass density give second-order effects. In Figure 8.35a and b, the transfer function and the ellipticity for a single layer are plotted. The peak of the horizontal ellipticity is close to the natural frequency. The cross-plot (Figure 8.35c) shows the relationship between the natural frequency and the peak ellipticity for a Monte Carlo sample of 2000 cases with a sublayered low-velocity layer over bedrock, with an impedance contrast larger than 3.5.

The ellipticity can be used to estimate relevant site properties, but measuring the ellipticity with active data requires multicomponent recording, which is not a standard practice. On the contrary, with passive data, some of the single-station multicomponent techniques are based on the estimation of the average spectral ratio.

8.5.1 Mode identification in the case of high velocity contrasts

One application of multicomponent data is the identification of modes, particularly for ambiguous cases. The mode osculation is the phenomenon where two modes get very close to each other, and their amplitude is also creating a smooth transition. It generally happens with large impedance contrasts. In Figure 8.36, the transition between the fundamental mode and the first higher mode is seamless and smooth for the velocity and the amplitude.

In this case, the energy jumps from the fundamental mode to the higher mode, creating an apparent continuous curve. Even in less extreme cases, the transition can be continuous. With the limited spectral resolution of real data, the identification of the mode superposition and transition may be impossible (see also Section 7.4). In Figure 8.37, the limited spatial resolution leads to an apparent dispersion curve that is associated with the fundamental mode in the high-frequency range and to the first higher mode in the low-frequency range (Boaga et al. 2013).

Misidentification can be avoided if the horizontal component is recorded. Indeed, the horizontal in-line component allows for identifying the fundamental mode below the osculation frequency as shown in Figure 8.38.

8.5.2 Passive H/V

The single-station passive methods, based on the analysis of the spectral ratio between the horizontal-to-vertical components, have been extensively debated and investigated. A general agreement on the important role played

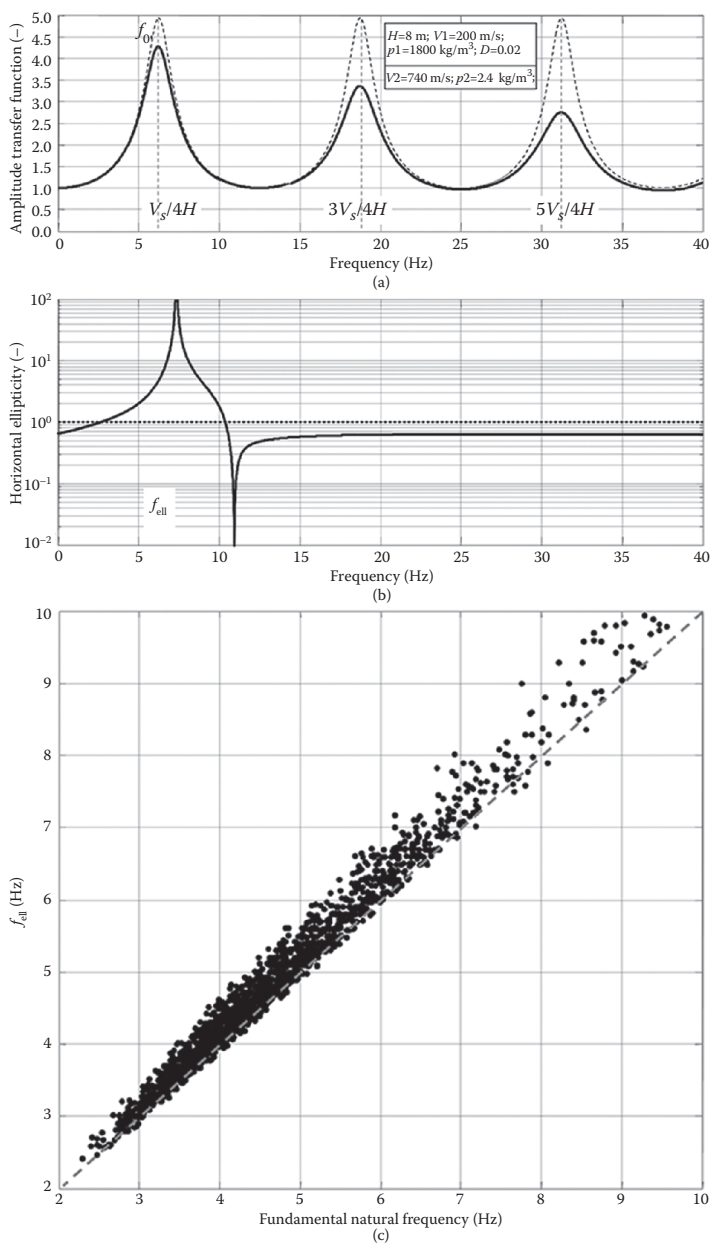


Figure 8.35 Horizontal ellipticity (H/V) of the fundamental mode: (a) amplification function for vertically propagating shear waves; (b) Rayleigh wave ellipticity; (c) comparison of the natural frequency with the frequency at which the maximum ellipticity is obtained.

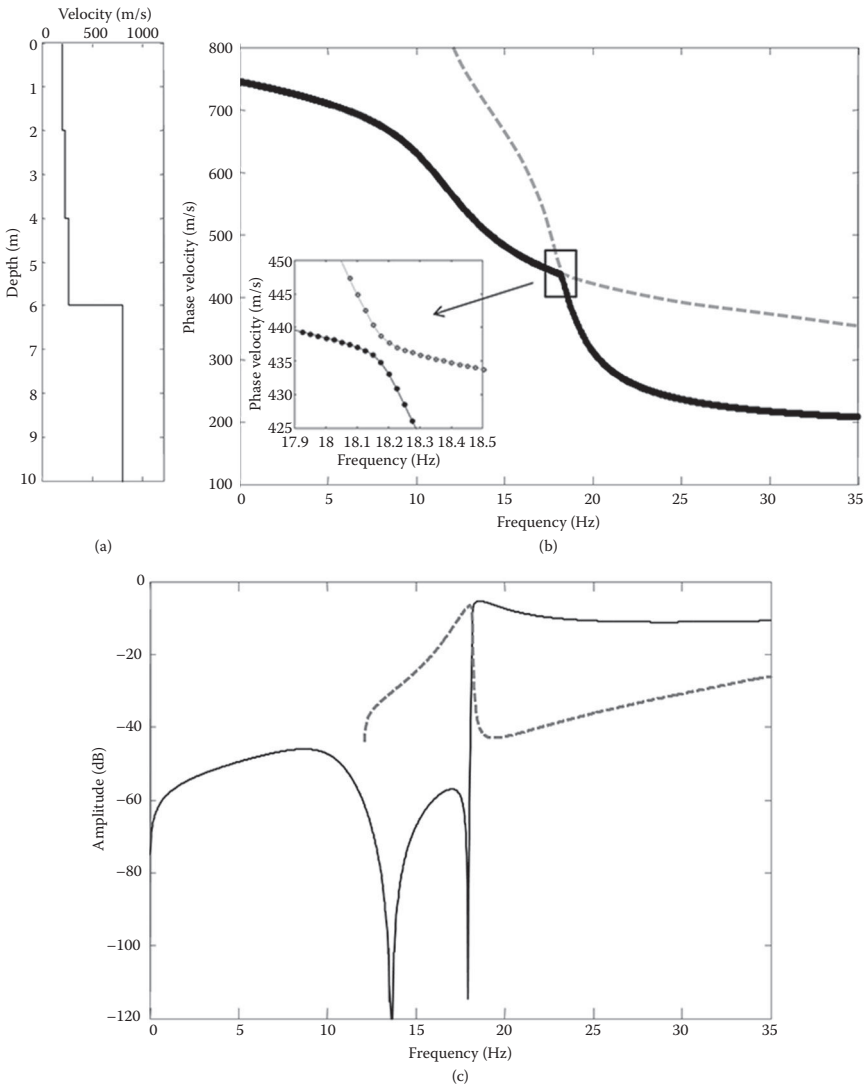


Figure 8.36 Synthetic example of model with an osculation point: (a) shear wave velocity model; (b) Rayleigh wave dispersion curve; (c) modal amplitudes for the first two higher modes.

by the Rayleigh wave ellipticity has been reached. Even if, in some cases, the role of body waves cannot be neglected, passive H/V measurements are related to surface wave propagation in most cases.

Ambient vibration techniques with single multicomponent stations and the H/V method can contribute effectively to site effect evaluation,

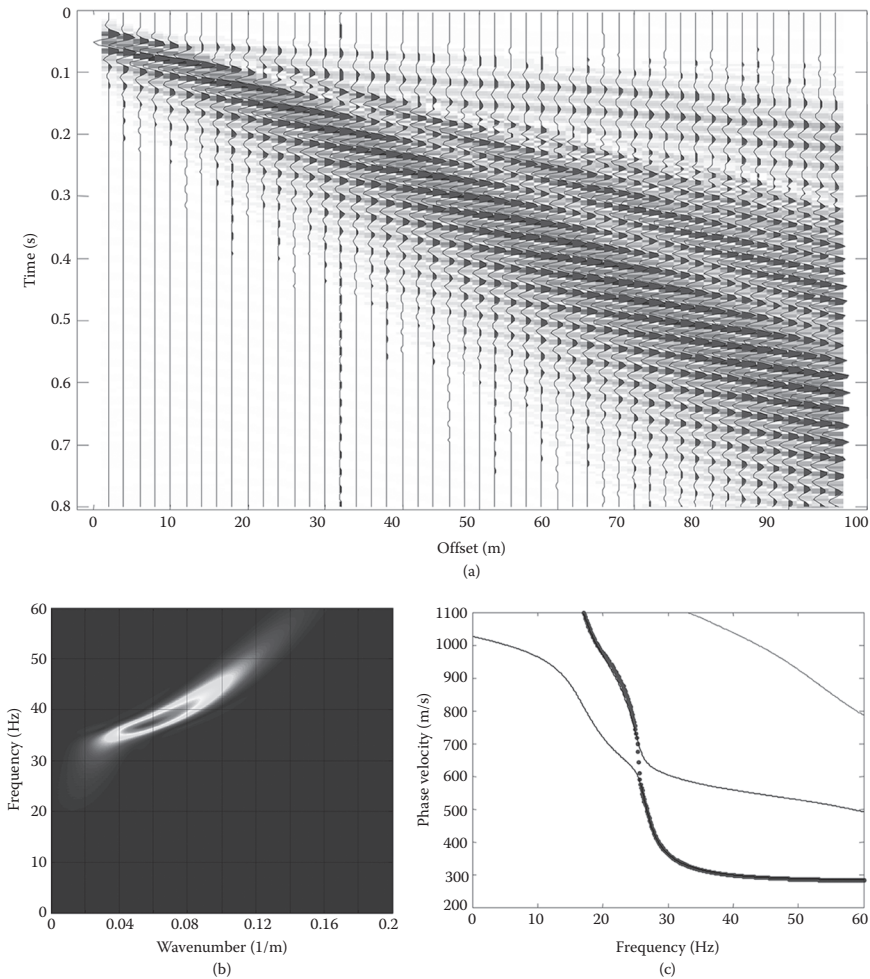


Figure 8.37 Synthetic data showing the effects of limited spatial resolution: (a) shot gather; (b) $f-k$ spectrum; (c) apparent dispersion curve from the maxima of the $f-k$ spectrum compared to the theoretical modal curves.

especially in seismic hazard microzonation, and particularly in urban areas. These low-cost techniques are also of great interest in moderate seismicity and in developing countries.

The H/V spectral ratio technique, based on the interpretation or inversion of the ratio between the Fourier amplitude of the horizontal and vertical components of microtremors, was introduced by Nogoshi and Igarashi in 1971 and became popular with the work by Nakamura (1989, 1996, 2000). Nakamura's theoretical explanation assumes an ambient noise wave field

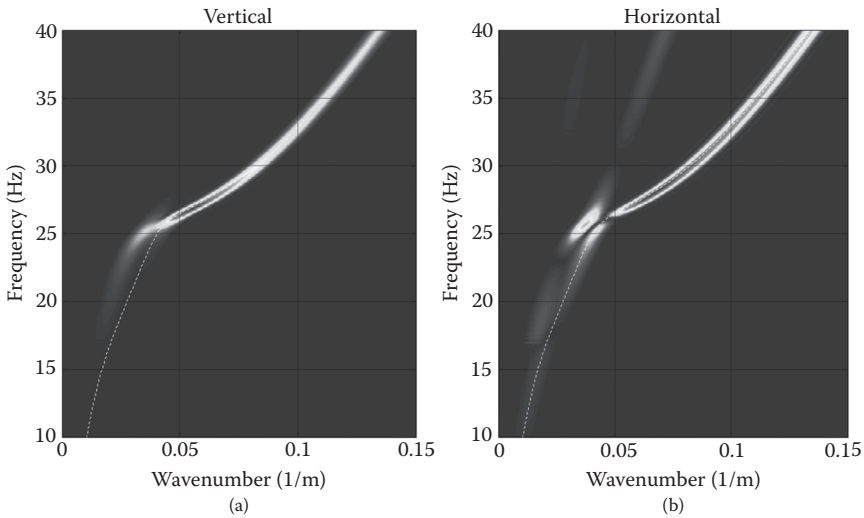


Figure 8.38 The f - k spectra for the same synthetic dataset of Figure 8.37: (a) vertical component; (b) horizontal component.

dominated by body waves, with deep sources, and explains the shape of the curve with the shear wave resonance in the sediments above bedrock. If the H/V curves are controlled by the shear wave resonance in the sediments, peak frequency and the amplitude depend on the soil transfer function and can be interpreted as the fundamental resonance frequency and the site amplification factor for low strain.

A large number of experiments and theoretical and numerical studies have demonstrated that, in most situations, the noise wave field is dominated by surface waves (Horike 1985; Arai and Tokimatsu 1998) and that the peak observed in the spectral ratio curve is related to the ellipticity of the Rayleigh waves.

When data are acquired and processed properly, the H/V ratio depends on the site properties. When the sources of the ambient noise are inside of the sedimentary layer or at the surface and close to the recording station, the peak is related to the horizontal ellipticity of the fundamental mode of the Rayleigh wave. A second peak related to the head shear wave can appear with far sources. With deep sources, in the bedrock, the peaks are due to the shear wave resonance of the head wave. The relative importance of the Rayleigh and Love waves can affect the shape of the curves, and this can vary from site to site.

8.5.3 How to compute the H/V

The acquisitions and processing of H/V data are discussed extensively in the documentation of the SESAME project (SESAME 2004), as well as

details for the identification of clear and reliable peaks. Low-frequency multicomponent receivers have to be used. The sensors and digitizer should guarantee sensitivity over the whole frequency range of interest, low electronic noise, and synchronized recording of the different components with the same system response.

The experimental conditions affect the measurement. Data should be collected avoiding underground structures, soft and irregular soil (mud, grass, ice, gravel, etc.), and some sources of noise such as machinery and traffic loads.

The duration of the records depends on the noise conditions and on the expected frequency of the peak—from a few minutes for shallow soil deposits to half an hour when the expected peak is at a frequency below 1 Hz. The duration should be increased if there are transient disturbances.

The data processing and the H/V computation involve several steps: selection of the stationary parts of data, rejecting transient pulses related to specific sources, computation of the Fourier spectra, smoothing of the spectra, merging of the horizontal components, computation of the H/V ratio, averaging of acquisition over different time windows, and estimation of average values and associated standard deviation to identify the reliable peaks.

8.5.4 Interpretation of H/V

In the original interpretation proposed by Nakamura (1989), the spectral ratio is directly related to the S-wave site response. However, the agreement in the seismological and engineering community is that the surface waves dominate the H/V response. With large velocity contrasts and with shallow sources close to the recording station, the role of the surface waves dominates. In some cases, with a noise wave field dominated by fundamental mode Rayleigh waves, not only can a clear large maximum be identified but a minimum may also be visible (related to the zero of the H/V ellipticity) such as in Figure 8.39.

In some cases, the shape of the curve indicates quite clearly which types of events have been recorded and analyzed; the curve can be interpreted as the response of the site to a certain phenomenon. The frequency of the H/V peak can be interpreted as the peak of the fundamental mode ellipticity and can be used to estimate the natural frequency of the site. In this case, the value of the peak is not related to the site amplification factor.

However, in general, there are different factors that can affect the shape of the H/V curve. The unknown nature and position of the seismic sources play a dominant role in determining which type of seismic waves compose the passive data, and consequently, this largely affects the H/V response.

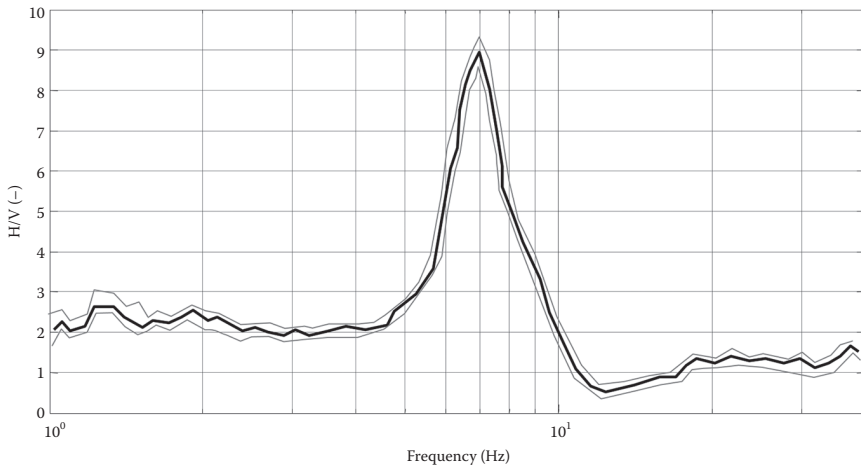


Figure 8.39 Example of H/V spectral ratio from experimental data.

The possible ambiguity and nonuniqueness of the computed curve should not be neglected. Using a single curve for the characterization of a site can lead to severe uncertainties.

The interpretation of the H/V curves requires a preliminary careful evaluation of their characteristics. As first criterion, the clarity of the peaks can be used. When the peak is clear (and does not have an industrial origin), it indicates the presence of a large impedance contrast at some depth, and the peak frequency is close to the fundamental frequency of the site. This is quite independent from the nature of the wave field. If the thickness is known, an estimation of the shear wave velocity can be obtained. If, on the contrary, the curve is flat (typically, values between 0.5 and 2), it is likely that the local velocity structure has no impedance contrasts. Different situations are possible, and detailed interpretation guidelines have been produced (SESAME 2004).

More advanced processing and interpretation techniques involve the inversion of the curve for the identification of model parameters as in surface wave dispersion inversion (Fäh et al. 2001). The curves are first corrected for the contamination by SH and Love waves and then inverted. Arai and Tokimatsu (2000) showed that H/V spectral ratios can be better reproduced if the contribution of higher modes of Rayleigh waves and Love waves is also taken into account. For the inversion, the strong non-linearity of the curves makes global search preferable, and the limited amount of information in the ellipticity curves requires the introduction of a priori information (e.g., on bedrock depth) or further constraints for

a reliable interpretation. In any case, it is important to take into account uncertainties related to solution nonuniqueness, which are very relevant.

Joint inversion of dispersion curves and H/V spectral ratios have been applied to increase the reliability with respect to each individual inversion (Parolai et al. 2005). Multichannel, multicomponent acquisitions allow the simultaneous extraction of dispersion curves and spectral ratios. With these approaches, both Rayleigh and Love waves can be identified and processed, better constraining the inversion.